From: "Journal of the Academy of Marketing Science" <em@editorialmanager.com>
Subject: **Decision on JAMS-D-17-00188R3**
Date: January 10, 2019 at 1:17:36 PM EST
To: "Carlos Bauer" <cbauer@cba.ua.edu>
Reply-To: "Journal of the Academy of Marketing Science" <jams.man.ed@gmail.com>

10 Jan 2019

Re: JAMS-D-17-00188R3, "The Impact of Superstar and Non-Superstar Software on Hardware Sales: The Moderating Role of Hardware Lifecycle"

Dear Dr. Bauer:

I have now completed my final review of your manuscript entitled "The Impact of Superstar and Non-Superstar Software on Hardware Sales: The Moderating Role of Hardware Lifecycle" and am pleased to accept it for publication in the Journal of the Academy of Marketing Science. I deeply appreciate the time and effort you have spent on preparing the manuscript, including your changes in this last round to shorten its overall length. The result is a fine contribution that I am sure will generate keen interest, debate, and additional research on the topic.

Your paper has now been sent to the publisher, Springer, for processing. They will be sending you page proofs in due course; at that time you will be asked to check the proofs and return them with corrections. Keep in mind that when you return your corrections, the paper will be published online (in advance of print publication), at which point no further changes can be made. So please be sure to check the proofs carefully.

For queries regarding your accepted paper, please click the following link http://www.springer.com/journal/11747; then click on "Contacts", and then "Production Editor", complete the query form and click "Submit".

Once again, thank you very much for your fine contribution; I hope you will continue to consider JAMS as a potential publication outlet for your future research.

Sincerely,

John Hulland
Editor-in-Chief
Journal of the Academy of Marketing Science

The Impact of Superstar and Non-Superstar Software on Hardware Sales: The Moderating Role of Hardware Lifecycle

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Abstract

We propose hardware lifecycle as a key moderator of the impact of superstar and non-superstar software on hardware adoption. A hardware's earlier adopters, are less price sensitive and have a higher preference for exciting and challenging software. In contrast, later adopters are more price sensitive and prefer simplicity in software. Superstars tend to be more expensive and more complex compared to non-superstars. Therefore, earlier (later) adopters prefer superstars (non-superstars), which leads to higher impact of superstars (non-superstars) on hardware adoption in the early (later) stages of the hardware lifecycle. Using monthly data over a twelve-year timeframe (1995 – 2007) from the home video game industry, we find that both superstar and non-superstar software impact hardware demand, but they matter at different points in the hardware lifecycle. Superstars are most influential when hardware is new and this influence declines as hardware ages. In contrast, non-superstar software has a positive impact on hardware demand later in the hardware lifecycle, and this impact increases with hardware age. Findings reveal that eventually the amount of available non-superstar software impacts hardware adoption more than the amount of available superstar software. The authors provide several managerial implications based on these findings.

Introduction

Two-sided markets provide value to consumers mainly through the pairing of a platform (e.g., Microsoft's Xbox One) and content that makes use of the platform (e.g., Titanfall) (Y. Lee and O'Connor 2003; Steiner et al. 2016). In the literature, the platform is typically referred to as hardware[1] while the content is typically referred to as software (Rysman 2009).[2] Two-sided markets are characterized by *indirect* network effects (Marchand and Hennig-Thurau 2013) wherein the value of hardware to its consumers increases as more software from content creators becomes available on the hardware. For example, consumers increasingly value smartphones (e.g., Apple Inc.'s iPhone and Samsung's Galaxy) as app developers create more mobile apps for these platforms. Similarly, in the video gaming context, users value a gaming console more as game developers create a wider selection of video games (i.e., software) for the gaming console (i.e., hardware). As consumers perceive higher value in a particular hardware due to wider availability of software, more consumers adopt the hardware, resulting in a larger group of consumers (known as an installed base) using the hardware. This self-reinforces the market, as a larger installed base encourages developers to create software for hardware with more end users (Church et al. 2008; Clements 2004; Rysman 2009).

While theory suggests that software availability creates a positive effect on hardware sales, two sets of empirical studies with complementary findings emerge. The first stream of literature focuses on the amount of available software in general and shows that a greater amount of available software increases hardware demand (Clements and Ohashi 2005; Dranove and Gandal 2003; Dubé et al. 2010; Gandal et al. 2000; Gretz and Basuroy 2013; Nair et al. 2004). A second stream of research examines the software – hardware demand link separately for different

[1] We acknowledge that certain types of software may act as platforms too, e.g. operating systems
[2] We follow this convention throughout the paper and refer to the platform as *hardware* and the content as *software.*

types of software. In particular, this literature categorizes software into superstars (i.e. software of exceptional quality) and non-superstars (i.e. the balance of software that may even include good-to-average quality products). These studies show that the positive effect of the amount of available software on hardware sales only exists for superstar software. Several studies in this stream of research document a positive impact of the amount of superstar software available on hardware demand (Binken and Stremersch 2009; Corts and Lederman 2009; Kim et al. 2014; R. S. Lee 2013).

Even in practice, examples abound where in some cases the amount of available software and in other cases the quality of particular software titles led to higher sales of hardware. For instance, the relatively mediocre performance of Microsoft's Xbox One gaming console compared to Sony's PlayStation 4 gaming console (Gilbert 2017) is often attributed to a limited amount of available games and media for Xbox One (Moorhead 2013), highlighting the importance of the amount of available software in the video gaming business. At the same time, introductions of superstar games on Xbox One and PlayStation 4 impacted their sales substantially. Sales of Xbox One doubled after the launch of the highly celebrated game Titanfall (Gaston 2014) and the launch of hit titles like FIFA 18 and Uncharted led to year-on-year revenue growth of 35.4% for Sony's PlayStation division (Kharpal 2017).

Although the previous literature has demonstrated the critical role superstars play in indirect network effects, the literature is largely silent on whether superstars matter for hardware demand throughout the hardware lifecycle. Similarly, the role of non-superstars in hardware adoption over the hardware lifecycle remains unexplained in the previous findings. As such the objective of this research is to extend this literature by studying how the effect of the amount of available superstar and non-superstar software on hardware sales varies over the hardware lifecycle, defined as the aging process the hardware experiences during its useable life.

According to the literature on diffusion of innovations (Rogers 2003), considerable heterogeneity exists among consumers over a hardware's lifecycle. Earlier adopters, who purchase the hardware during the initial stages of its lifecycle, are price insensitive and seek challenge and excitement from engagement with new technology. Conversely, later adopters tend to be price sensitive and prefer simplicity in technology usage (Jahanmir and Lages 2016). Generally, superstar software tends to be more expensive and more complex than non-superstar software. Therefore, due to the heterogeneity in their preferences, earlier adopters are more likely to prefer superstar software and later adopters are more likely to prefer non-superstar software. This suggests that for earlier adopters, the amount of available superstar software increases the perceived value of hardware more than it does for later adopters. In contrast, the amount of available non-superstar software increases the perceived value of hardware more for later adopters than it does for earlier adopters. Thus, we expect that the amount of available superstar software will have a higher (lower) impact on hardware demand in the earlier (later) stages of the hardware's lifecycle. Conversely, the amount of available non-superstar software will have lower (higher) impact on hardware demand in the earlier (later) stages of the hardware's lifecycle.

We test the implications of our theory by utilizing a comprehensive dataset (1995 – 2007) on monthly console sales and game supply from the U.S. home video game industry. We develop a systems of equations econometric model which incorporates the essential feedback loop in indirect network industries, where hardware demand is a function of software supply, and software supply is a function of hardware installed base (Kim et al. 2014; Nair et al. 2004). We expand on previous models by considering the feedback loop for different software types (superstar and non-superstar), rather than an aggregate software supply equation.

We report several interesting findings. By considering the effects of software types on hardware demand, we find that both superstar and non-superstar software impact demand, but

they matter at different points in the hardware lifecycle. Superstars are most influential when hardware is new, but the disproportionately large effect that superstar software exerts on hardware purchases cannot be generalized throughout the life of the console, and it is limited to newer hardware. In contrast, non-superstar software has a positive impact on hardware demand later in the hardware lifecycle, and this impact increases with hardware age. Findings reveal that eventually the amount of available non-superstar software impacts hardware adoption more than the amount of available superstar software.

Furthermore, we find that the effect of hardware installed base on software supply differs significantly depending on the type of software. For example, the elasticity of superstar software supply to installed base varies across the life of the hardware, while the elasticity of non-superstar software supply is relatively constant and significantly larger than that of superstars. For non-superstars our results are similar to Stremersch, Tellis, Franses, and Binken (2007) who find that hardware sales influence software sales and not vice-versa early in the product lifecycle. However, our results are the opposite for superstars. We find that the elasticity of superstar supply to installed base is low early in the hardware lifecycle, around the time when the impact of superstar supply on hardware demand is greatest.

This research makes several contributions to theory and practice. First, by identifying the age of hardware as a critical moderator of the effects of the amount of available superstar and non-superstar software on hardware demand, we establish boundary conditions on previous findings in the literature – both superstars and non-superstars matter, but at different point in the hardware lifecycle. Importantly for theory, we find the hardware lifecycle influences the impact of key strategic variables (e.g. software variety and quality) on hardware adoption. While our focus is limited to the impact of different software types, it is likely that the impact of other common strategic variables in the marketer's toolbox on both the hardware and software side

(e.g. price, advertising, and distribution) vary with the hardware lifecycle. Thus, this research provides guidance for studying additional limiting conditions for the effectiveness of different marketing strategies over the hardware lifecycle in platform and two-sided markets.

Second, we highlight that in different phases of its lifecycle, hardware attracts different types of consumers. We argue that consumer heterogeneity leads to different preferences for superstar and non-superstar software, and consequent hardware demand. The failure to account for heterogeneity among the earlier and later hardware adopters may lead to incorrect estimation of the impact of software types on hardware demand.

Third, for practitioners, our results help clarify the role of non-superstar software in extending the life of hardware, an area of research that has received scant attention despite its importance for practitioners. Marketing product innovations in declining stages of product lifecycle is a highly challenging task for managers (Marchand 2016). Our results suggest that wider availability of non-superstars will help extend the life of hardware.

Fourth, our findings speak to the strength of the indirect network effect over the hardware lifecycle. The strength of the indirect network effect is characterized by the intensity of the feedback loop where more hardware adopters result in more software supply which, in turn, results in more hardware adopters. We find that the impact of hardware installed base on superstar software supply is relatively weak when hardware is new. Importantly, this effect occurs at precisely the time when the amount of available superstar software has the greatest impact on hardware adoption. Put together, this suggests that the indirect network effect is relatively weak early in the hardware lifecycle. This has implications for software supply strategies. Hardware managers should consider compensating superstar software developers for

this weaker indirect network effect early in the hardware lifecycle via specific strategies like bundle offers, coupons, subsidized advertisement, and subsidized development[3].

On the other hand, our evidence suggests the strength of the indirect network effect grows later in the hardware lifecycle. This is because the impact of hardware installed base on non-superstar software supply remains relatively constant as hardware ages while the effect of the amount of available non-superstars on hardware demand becomes stronger. As hardware matures, hardware managers can enhance the stronger indirect network effect by incentivizing the provision of more non-superstars relative to superstars.

Finally, our results have implications for structuring compensation policies in the video game industry. Game publishers often tie bonuses of game developers to the availability of superstars. However, hardware manufacturers can offer separate incentives to game developers for making simpler and less expensive games in the later stages of hardware lifecycle.

Theory and Hypotheses Development

Network effects exist when consumers benefit either directly or indirectly as more consumers purchase the same product. Direct network effects are present when the number of adopters of a network directly impact the utility of consumers (Katz and Shapiro 1994). The classic example is an online social network such as Facebook wherein each consumer's benefit of being on the social network increases as more consumers adopt the social network, leading to more opportunities to socialize. Indirect network effects exist when a consumer's utility increases because greater adoption of a network causes an increase in the number of products for the network (Clements 2004; Rysman 2009). Indirect network effects underlie the typical hardware and software relationship. For example, more consumers purchasing the same hardware incentivizes software developers to make more software available for that hardware. A

[3] We thank an anonymous reviewer for this suggestion

wider variety of available software in turn makes the hardware more attractive to consumers. This positive feedback loop associated with indirect network effects has received extensive attention in the theoretical literature (e.g., Church et al. 2008; Clements 2004; Hagiu and Wright 2015; Rochet and Tirole 2006) with an increase in attention to strategic considerations for both software and hardware firms (For a summary see McIntyre and Srinivasan 2017).

The empirical literature mainly focuses on quantifying the impact of the feedback relationship embedded in indirect network effects on market outcomes, specifically focusing on the influence of increased software availability on hardware sales and market share. Broadly, two major streams of research emerge from the extant literature. The first stream of literature focuses on the software availability in general. A common theme in this literature stream is that the amount of available software positively influences hardware demand. At the same time, hardware installed base positively impacts software supply. For a brief review of the literature, see Table 1. Several studies show that regardless of software quality, the amount of available software is an important determinant of hardware success, and that hardware installed base drives the amount of available software (e.g., Nair et al. (2004) in the personal digital assistant industry; Clements and Ohashi (2005), Dubé et al. (2010), and Gretz and Basuroy (2013) in the video game industry; Gandal et al. (2000) in the compact disc industry; Dranove and Gandal (2003) in the digital video disc industry). However, in a study spanning nine industries, Stremersch et al. (2007) find that hardware sales lead software sales but the reverse almost never happens. Stremersch et al. (2007) focus exclusively on the impact of software on hardware (and vice versa) when new technology is introduced (not over the entire product lifecycle) and do not consider the impact of different types of software (i.e. superstar or non-superstar) on hardware sales. Collectively, these findings suggest that there are unexplored contingencies that may explain varying impacts of indirect network effects reported in the extant literature.

-- Insert Table 1 about here --

The second stream of literature goes beyond the amount of available software in general and focuses on the amount of available software of different qualities. Therefore, this literature stream considers software quality as a moderator of the impact of indirect network effects on hardware demand. Extending the work of Stremersch et al. (2007), Binken and Stremersch (2009) consider the impact of different *types* of software on hardware sales, and find that the introduction of superstar software, as opposed to non-superstars, drives hardware sales. As the focus of Binken and Stremersch (2009) is on assessing the impact of different software types (superstar and non-superstars) on hardware demand, they did not model the supply of superstars and non-superstars as a function of hardware sales. Continuing this line of research and incorporating software supply estimations, Corts and Lederman (2009) and Kim et al. (2014) find that all software exert an influence on hardware sales, but hit titles are significantly more important than those that are merely average. More recently, Kretschmer and Claussen (2016) demonstrate that software compatible with previous generation hardware significantly drives demand for new-generation hardware early in the hardware's lifecycle.

Although the previous literature has demonstrated the critical role superstars play in indirect network effects, the literature is largely silent on whether superstars matter for hardware demand throughout the hardware lifecycle. Similarly, the role of non-superstars in hardware adoption over the hardware lifecycle remains unexplained in the previous findings. In two-sided markets, a product's aging process has a significant effect on the economics of its complement (Day 1981; Golder and Tellis 2004). The above-cited literature typically controls for time-related trends in their estimations and the typical finding is the unsurprising result that, holding everything else constant, hardware sales tend to decrease with hardware age. However, no previous study considered how the hardware lifecycle may moderate the impact of the number of

available superstars and non-superstars on hardware sales. A careful review of the literature produced only a limited number of studies that considered the moderating effects of hardware lifecycle in any empirical setting (e.g., Gretz and Basuroy 2013; Marchand 2016; Sun et al. 2016). Our theoretical framework suggests that both superstar and non-superstar software will influence hardware adoption, but the relative impact of each software type on hardware demand will change over the hardware lifecycle. Figure 1 summarizes the discussion above and identifies the research gap we aim to fill.

-- Insert Figure 1 about here --

Below we discuss superstars in general as well as some key characteristics of superstar software specific to the video game industry. We then discuss consumer heterogeneity over the product lifecycle to generate hypotheses concerning when superstar and non-superstar software are likely to be most influential on hardware adoption.

Superstars and Superstar Games

The phenomenon of superstars pertains to relatively small numbers of entities, which earn enormous amounts of money and dominate the activities in which they engage (Rosen 1981). Superstar effects have been studied extensively in entertainment and sports. For example, Basuroy, Chaterjee, and Ravid (2003) find that superstar actors may dampen the impact of negative critical reviews on movie box office. Along the same lines, in a recent meta-analysis Hofmann et al. (2017) observe that an inclusion of a commercially successful star actor in a movie results in $12.46 million more in box office revenues. Hausman and Leonard (1997) show that National Basketball Association ratings are significantly higher when superstar players are involved and the benefits spillover to teams without superstars when they play teams with superstars. Similarly, Gooding and Stephenson (2017) find that ratings for certain tournaments went up 50% if superstar golfer Tiger Woods was in contention.

Superstars are gaining increasingly more attention in the video game industry literature since Binken and Stremersch (2009) showed that console makers disproportionately benefit from superstar video games released on their platforms. As Table 1 shows, several studies find that the indirect network effects that help console makers are almost all due to superstar games. Binken and Stremersch (2009), along with other researchers (e.g., Kim et al. 2014) categorize games as high quality or superstars if they cross a threshold composite critical review score which is usually quite high. For example, Binken and Stremersch (2009) use a cutoff of minimum 90 out of 100 composite score from third-party critics, which very few games achieve. We note that this is not influenced by the game's revenue as critic scores are released *before* the game. However, games that achieve this critical threshold tend to be much more successful commercially than the games that do not (Binken and Stremersch 2009; Sun et al. 2016). Importantly, those games that achieve this rarified level of critical acclaim have other features in common in addition to a greater likelihood of commercial success. First, superstar games tend to be more engrossing, challenging and complex to play, and second, superstar games are usually more expensive than other games.

Superstar games tend to be more challenging to play because of greater in-game complexities, which leads to higher engagement especially for more serious players. A few industry observers note that high quality games are challenging and force players to make decisions at each stage (Totilo 2012). Squire and Jenkins (2002, p. 70) note that "well-designed game environments present players with clear goals,...these games create 'possibility spaces,'…that provide compelling problems…and then respond to players' choices with meaningful consequences." Indeed, engagement or play engrossment is an important determinant of game users' satisfaction with experience (Phan et al. 2016). Pearce (2002) finds that game players do not like to be spoon-fed their narrative and highlights the importance of leveraging

narrative to increase engagement. Lack of challenge in gameplay or poor narrative typically leads to poor critical review scores (Bernbeck 2015).

Additionally, superstar games are on average more expensive for end users than other games. In our data the average price of a superstar game ($52.83) is 27.01% greater than the average price of a non-superstar ($41.59).[4] More importantly, as shown in Figure 2, the average price of superstar games has been consistently more than that of non-superstar games in our study period from 1995 to 2007, with the exception of only 2006.

-- Insert Figure 2 about here --

Two primary potential reasons for high prices exist. First, due to their high quality, the development costs associated with superstar games are likely to be high. We conjecture that even after console makers subsidize part of these costs, game developers will pass on the rest of the costs to consumers, leading to higher prices. However, independent of development costs, another plausible reason for high prices is that prices act as signals of quality (Zeithaml 1988) especially for experiential goods such as video games. Therefore, game developers may signal high quality with high prices for superstar games.

Earlier and Later Adopters of Hardware

Innovation diffusion literature contends that considerable heterogeneity exists among the adopters of an innovation over the innovation's lifecycle (Rogers 2003). Although Rogers (2003) proposes five discrete segments of adopters—innovators, early adopters, early majority, late majority, and laggards, our theory simplifies this into two categories—earlier adopters, who embrace an innovation in the earlier stages of its lifecycle and later adopters, who embrace an innovation in the later stages of its lifecycle.

[4] These prices are significantly different at $p < 0.01$ using a standard T-test. Average price calculated for each game adjusting for inflation using the Consumer Price Index for Urban Consumers (2015 = 100).

This way of categorizing innovation adopters has two benefits. First, such categorization does not rely on a rigid definition of what constitutes a certain type of adopter because for any given point in time the consumers who have already adopted the innovation can be considered "earlier adopters" and the consumers who are yet to adopt the innovation can be considered "later adopters." For our theory the sufficient condition is that earlier adopters are different from later adopters on some important dimensions, which we discuss below. Second, this categorization helps us in applying empirical generalizations from related literature that may have used adopter segments different from what Rogers (2003) proposed.

Past research has documented several differences in socioeconomic characteristics, personality variables, and communication behaviors pertaining to earlier and later adopters (Rogers 2003). In our theoretical framework, we consider differences along the following two dimensions: 1) price sensitivity and 2) preference for simplicity. We argue that each of these dimensions influences consumers' willingness to purchase a superstar game, which in turn affects console adoption.

Price sensitivity. Prior research shows that earlier adopters have higher social status relative to later adopters (Rogers 2003). Social status is typically measured in terms of income, possession of wealth, standard of living, etc. As such, earlier adopters are likely to be less price sensitive and can afford more expensive superstar games. Additionally, earlier adopters of an innovation tend to value that innovation more highly than later adopters (Rietveld and Eggers 2018). This further implies that earlier adopters will be less price sensitive and more willing to purchase an expensive superstar game. On the opposite end of the spectrum, later adopters are both less financially able and less willing to value innovation at higher levels. This implies that later adopters will be more price sensitive and less willing to purchase an expensive superstar game.

Preference for simplicity. Later adopters have a higher preference for simplicity (Jahanmir and Lages 2015, 2016). Furthermore, later adopters have difficulties in dealing with complex ideas due to their abstraction (Rogers 2003). Jahanmir and Lages (2015) note that "while innovators identified more sophisticated needs and problems (e.g. more options on laptops or more applications for smartphones), *laggards seek simplicity and convenience*[5] (e.g. fewer programs on a laptop, simplified iPhone)." Similarly, Jahanmir and Lages (2016) report that "users who are more skeptical and resistant to innovations prefer products that are simple to use." As superstar games tend to be more complex and challenging, later adopters may prefer them less compared to non-superstar games which are relatively simpler. On the other hand, compared to later adopters, earlier adopters have higher aspirations (Rogers 2003), which, we argue, will lead to a higher preference for superstars as they are more challenging and complex. In addition, Reitveld and Eggers (2018) suggest that earlier adopters seek novelty and therefore prefer superstars over non-superstars. In contrast, later adopters avoid ambiguity and therefore are more likely to prefer non-superstars as they are less ambiguous compared to superstars (Rietveld and Eggers 2018).

We note, however, that later adopters are generally more risk averse than earlier adopters (Rogers 2003). Due to their higher risk aversion, later adopters might prefer superstars more than non-superstar games because superstar games have higher critical ratings, implying less uncertainty about the quality of the game. However, previous literature also suggests that later adopters are less likely to do extensive research and their search behavior is different from earlier adopters (Rietveld and Eggers 2018). While superstars may appear less risky to later adopters due to their better quality, such consumers are also less likely to have a concrete idea about the high quality of superstars because they do not search for information extensively. Furthermore,

[5] Emphasis ours

we argue that higher price and higher complexity will deter later adopters from purchasing a superstar even though they may perceive it as less risky in terms of quality.

Lower price sensitivity and lower preference for simplicity of earlier adopters suggests that superstars will lead to higher purchase of gaming consoles early in its lifecycle. As the gaming platform ages, the target market becomes more populated by the type of later adopters who tend to show higher price sensitivity (Clements and Ohashi 2005) and higher preference for simplicity. Their higher price sensitivity suggests that later adopters will not be interested in purchasing superstars which are usually more expensive than other games. Similarly, superstars tend to be more complex and challenging, which is an undesirable property in games for later adopters who prefer simplicity. In sum, compared to earlier adopters, later adopters are less likely to buy superstar games and therefore adopt the gaming console because of the amount of available of superstar games. Thus, as a gaming console ages, superstars are less likely to lead to more console sales. This leads to the following two hypotheses:

H1: The impact of superstar games on the sales of compatible gaming consoles is negatively moderated by the age of the consoles such that superstar games lead to higher (lower) console sales in the early (late) stages of the console lifecycle.

H2: The impact of non-superstar games on the sales of compatible gaming consoles is positively moderated by the age of the consoles such that non-superstar games lead to higher (lower) console sales in the late (early) stages of the console lifecycle.

Next, we describe our data and empirically test the effects of software assortment on hardware sales and the moderating role of hardware lifecycle.

Data and Empirical Model

Our goal is to analyze superstar and non-superstar software effects as hardware ages. The US video game industry is ideal for this analysis for numerous reasons. First, the home video

game industry is often cited as a canonical example of platform-based systems (Corts and Lederman 2009; Dubé et al. 2010). Second, extant research shows that superstar games exert a significant influence on console market-share growth (Binken and Stremersch 2009; Cox 2014; R. S. Lee 2013). Third, total industry sales in console generations follow unique patterns over their lifespan (Gretz and Basuroy 2013), allowing us to study dynamic effects of superstar and non-superstar software as consoles age. Lastly, the video game industry exemplifies a context where indirect network effects play an important role as more software available makes the console more desirable to consumers, which motivates software developers to generate even more games. Our data allows us to model this feedback loop and obtain relevant managerial insights.

Our dataset comes from the NPD Group, a marketing research firm, and includes monthly point-of-sale data on prices and quantities for video game consoles in the US from January 1995 through October 2007. The dataset contains monthly average price and quantity for 15 distinct consoles, 923 observations overall, over four console generations. In addition, the dataset includes titles, introduction dates, unit sales, and revenues for all individual games released for each console over time. This includes games that are exclusive to one console, as well as those available on multiple consoles. Similar to other comparable studies, we consider the impact of all games available for a console on hardware demand (Clements and Ohashi 2005; Gretz and Basuroy 2013). We observe 5,044 unique games in the dataset resulting in 7,424 game/console observations, as several games are released on multiple consoles.[6] We provide the number of exclusive games[7] and total games available on each console, each console's release

[6] We note that games are physically distinct from consoles in that no console is preloaded with gaming software.
[7] While we do not differentiate between exclusive and non-exclusive games in the main econometric analysis in the paper, we provide similar analysis using only exclusive games and non-exclusive games in Appendix B.

date, the number of observations of each console in the dataset, and summary characteristics for each console generation in Table 2.

-- Insert Table 2 about here --

Superstar classification

Binken and Stremersch (2009) note that superstars in the video game industry display "increasing returns because of the scarcity of high quality" (p. 89). We follow the literature and leverage expert ratings to obtain a proxy for game quality for each game in the dataset. As Zhu & Zhang (2010, p. 139) note "products receiving positive reviews tend to be of high quality." As such, for our empirical tests, the classification of superstars should be based on quality rating and not game sales[8]. First, our theoretical framework highlights the importance of software complexity as a key characteristic that will influence consumer adoption patterns over the console lifecycle. The operationalization of superstars using critical acclaim captures this dimension because game complexity and user engrossment are typically what critics look for in a game (Bernbeck 2015). Second, while a classification based on game sales might alleviate the concern of highly rated games flopping in the market or low rated games achieving success, game sales are not known to managers before release. We want our analysis to be useful to managers based on the information they have before a game is introduced in the market.[9]

Our approach to operationalizing superstars is consistent with the considerable amount of literature on experiential goods that use expert ratings as indicators of quality in movies (Basuroy et al. 2003; Liu 2006), theatre (Reddy et al. 1998), music (Papies and van Heerde 2017), and video games (Binken and Stremersch 2009; Marchand 2016; Zhu and Zhang 2010). We obtained review data for each game in our dataset from Gamespot.com, a game-focused website reporting

[8] Though we note below that our results are robust to classifying games as superstars according to sales criteria.
[9] For face validity, we identify games in our dataset that sold 1 million copies or more. We find the average quality of these games (81.77) is significantly higher than the average quality of games that do not achieve this threshold (58.44). These averages are significantly different with > 99% confidence using standard t-test.

three summary ratings for each game: (1) Metacritic score, (2) User Average, and (3) Gamespot score. Whereas Gamespot score is based on a review from a Gamespot.com editor, User Average is the average of amateur reviews from Gamespot.com members. In contrast, Metacritic score is weighted average of many expert reviews aggregated by Metacritic.com, which curates a large group of the world's most respected critics and creates a weighted average to summarize the range of their opinions. Metacritic.com reports one single number called Metascore that captures the essence of critical opinion.

We use Metascore as the operationalization of our quality measure[10] for two reasons. First, critics tend to be less influenced by sales performance than amateur reviewers.[11] As a large majority of critic ratings are released before a game is out (Zhu and Zhang 2010), the Metascore for a game does not vary a lot over the life of the game. Indeed, Metacritic.com takes steps to ensure that expert reviews cannot be altered.[12] As such, Metascore is an ex-ante proxy of quality (i.e., before sales are observed) and useful for managers making decisions before a game is released. Second, Metascore is highly relevant for game publishers and game developers because typically publishers pay bonuses to developers based on Metascore (Usher 2015). Schreier (2015) reports that bonuses in the gaming industry are tied to achieving a minimum Metascore of 85 to 95.

Metascore ranges from a low of 0 to a high of 100, where a higher score is better. We adopt the dichotomous criterion used by Binken and Stremersch (2009) and classify games as superstars if their Metascore is at least 90, non-superstars otherwise. Games that achieve higher Metascore typically do better in the market. To illustrate this, Figure 3 shows the superstar and

[10] Though we note below that our results are robust to using different quality ratings to classify superstars.
[11] Our concern is that quality ratings by users may be influenced by sales performance, given that users enter their ratings after the game is released. As a robustness check, we also used a quality measure based on the average of the three ratings and found similar results to those presented below.
[12] Metacritic touts this policy on their website (http://www.metacritic.com/faq) as a way to protect critics from outside influences that may pressure them to change their scores.

non-superstar revenues per game for 20 different quality categories, with games in the top two categories classified as superstars.[13] Figure 3 shows clearly increasing returns in the top two quality categories. In our sample only 3.49% (259 out 7,424) of all games/console releases can be classified as superstars, yet they account for 16.62% of game unit sales and 18.61% of game revenue. This indicates that the games we classify as high quality or superstars are rare and they earn a disproportionate amount of sales and revenue relative to other games.[14]

-- Insert Figure 3 about here --

Estimation Strategy

For estimation, we allow the effect of predictor variables to change as the hardware ages. Furthermore, we jointly estimate hardware demand, non-superstar software supply, and superstar supply using a system of equations framework.

Feedback loop. In order to capture these hardware-software dynamics, we jointly estimate both the hardware demand side (where the number of games is included as an independent variable) and the software supply side (where the number of installed base consumers on a console is included as an independent variable) as a system of equations. Considering this process is imperative as failure to incorporate this feedback loop can bias estimates (Clements and Ohashi 2005; Gretz and Basuroy 2013; Kretschmer and Claussen 2016). Finally, jointly modeling the game supply and hardware demand side, as we will show, yields rich managerial insights, which would not be apparent in a reduced form model where installed base is included directly as an independent variable in the hardware demand specification. We begin by discussing the estimation equations, then briefly discuss endogeneity concerns, and review results.

[13] This is generated in a manner similar to Figure 1 in Binken and Stremersch (2009).
[14] We perform a series of robustness checks using different quality ratings to classify games as superstars. All estimations from robustness checks are available from the authors upon request.

Hardware demand. We derive hardware demand using a structural discrete choice model for differentiated products as in Berry (1994) and Berry et al. (1995). In each period *t*, consumers maximize utility by choosing to purchase (or not) one of the available consoles (i.e. the hardware) on the market. Formally, we define the utility consumer *i* receives from purchasing hardware *k* in period *t* as

$$u_{i,t}^{k} = \alpha_0 + \beta_p \text{Price}_t^k + \beta_{NS}\text{NonSuper}_t^k + \beta_{NSA}\left(\text{NonSuper}_t^k \times \text{Console Age}_t^k\right) + \beta_S \text{Super}_t^k$$
$$+\beta_{SA}\left(\text{Super}_t^k \times \text{Console Age}_t^k\right) + \beta_{Back}\text{Back}_t^k + \beta_{BackSuper}\text{Back Super}_t^k + Controls + \xi_t^k + \varepsilon_{i,t}^k$$

where α_0 is a constant, the βs are the marginal impact of observable hardware characteristics on utility, NonSuper_t^k and Super_t^k are respectively the number of non-superstar and superstar games available on console *k* at time *t*. Console Age_t^k is the age of console *k* in months at time *t* since it was first introduced in the market. *Controls* are additional console and industry control variables (described below), ξ_t^k is the impact of unobservable console *k* characteristics in period *t* on utility, and $\varepsilon_{i,t}^k$ is the individual specific shock for console *k* in period *t*. Following Berry (1994), we assume $\varepsilon_{i,t}^k$ is individually and identically distributed according to the double exponential and the utility of not purchasing in period *t* is normalized to 0. Additionally, we assume consumers are homogenous which results in the standard logit model for cumulative market share data, where the dependent variable is the difference between the natural log of console market share and the natural log of the outside option of not purchasing.[15] The estimation equation for hardware demand is therefore:

[15] The outside option is the fraction of people in the market per period who do not buy any console. The market in any period is defined as the number of US homes with TV sets who have not purchased a video game console. The number of US homes with a TV set was gathered from The Nielsen Company (http://en-us.nielsen.com/). We estimate the number of households that do not own a video game console from our data – in each generation we calculate total generational sales up to the period in question and subtract from the number of US homes with a TV set. Importantly, we depreciate generational sales by 90% per year to account for the fraction of consumers who

$$\ln\left(\text{Share}_t^k\right) - \ln\left(\text{Share}_t^0\right) = \alpha_0 + \beta_p \text{Price}_t^k + \beta_{NS}\text{NonSuper}_t^k + \beta_{NSA}\left(\text{NonSuper}_t^k \times \text{Console Age}_t^k\right) + \beta_S \text{Super}_t^k + \beta_{SA}\left(\text{Super}_t^k \times \text{Console Age}_t^k\right) + \beta_{Back}\text{Back}_t^k + \beta_{BackSuper}\text{Back Super}_t^k + Controls + \xi_t^k \quad (1)$$

where Share_t^k is the market share of console *k* at time *t*[16], Share_t^0 is the market share of the outside option at time *t*, and ξ_t^k is econometric error term.[17]

A number of consoles are backward compatible. That is, they can play games intended for older consoles. For example, PlayStation 2 can play games designed for the original PlayStation. Kretschmer and Claussen (2016) show that backward compatible consoles benefit from the supply of games available for the relevant previous generation console, though this is moderated by the age of the new console and its technological improvement over its previous generation counterpart. We account for backward compatibility with the inclusion of Back_t^k, which is the cumulative number of games released for previous generation consoles compatible with console *k* by time *t,* and Back Super_t^k, which is the share of these games that are superstars.[18] For example, the hardware demand equation for Sony PlayStation 2 will include the

purchase multiple consoles in each generation. In other words, we assume consumers can only buy one console a month, but some fraction returns to the market to make another purchase later. This approach has been used in Clements and Ohashi (2005), Gretz (2010), and Kretschmer and Claussen (2016). We should mention that our results below are robust to various depreciation rates for total generational sales.

[16] Market share of the focal console is calculated as the sales of the focal console relative to the total sales of all consoles in the same console generation as the focal console. Sales of previous versions of the console are not included in the market share calculation of the focal console.

[17] Using the logit demand model derived from the structural approach to model hardware demand has the advantage of controlling for the effect of competing product characteristics in a tractable way without entering them directly into the estimation equation (McFadden 1973). Otherwise, as Berry (1994) notes "a system of *N* goods gives N^2 elasticities to estimate" in order to account for competitive effects, which quickly becomes impractical to estimate as degrees of freedom decrease quadratically with the addition of competing products. However, as a robustness check we estimate the model for hardware demand using the natural log of console sales for console *k* in period *t* as the dependent variable and obtain qualitatively similar results to those presented below. These results are available from the authors upon request.

[18] We thank an anonymous reviewer for suggesting this operationalization of backward compatible superstars. While we obtain similar results to those presented below using the number of backward compatible superstars (rather than the share), this variable was highly correlated with the number of backward compatible games which lead to collinearity concerns in our estimations. Using the share of backward compatible superstars alleviates this issue.

cumulative number of games released for Sony PlayStation in Back_t^k and the share of those games that are superstars in Back Super_t^k. In general, we expect the coefficients on these variables to be positive as in Kretschmer and Claussen (2016), however, other research incorporating the impact of backward compatible games has had mixed results (Gretz 2010; Gretz and Basuroy 2013).

We also include Price_t^k, the price of console *k* at time *t*, and expect a negative and significant estimate of β_P, as price should have a negative impact on console utility. Finally, we include console age, month dummies, year dummies, and console dummies as console and industry control variables. Console age will capture any decrease in the attractiveness of a console as it ages (Clements and Ohashi 2005), month dummies will capture seasonal effects (i.e., holiday purchases), year dummies will capture any systematic demand changes from year to year, and console dummies will capture any time-invariant idiosyncratic console effects, such as console quality.

Software supply. Gretz (2010) observes that the natural log of the number of available games depends linearly on the natural log of installed base, where installed base is defined as the cumulative amount of monthly console sales preceding the observed period. For instance, a console's installed base observed in its 10th month will be the cumulative number of console sales up to and including the 9th month. While we do not theoretically model the supply of superstar and non-superstar games, we assume supply follows a similar process for all games found in Gretz (2010). We use the same predictors to estimate the supply of non-superstar and superstar games and estimate the following reduced-form equations:

$$\begin{aligned}\text{NonSuper}_t^k = c_0 + g_{IB}\text{IB}_t^k + g_{IBA}\left(\text{IB}_t^k \times \text{Console Age}_t^k\right) + g_{Back}\text{Back}_t^k \\ + g_{BackSuper}\text{Back Super}_t^k + Controls + \eta_t^k\end{aligned} \tag{2}$$

and

$$\begin{aligned}\text{Super}_t^k = d_0 + s_{IB}\text{IB}_t^k + s_{IBA}\left(\text{IB}_t^k \times \text{Console Age}_t^k\right) + s_{Back}\text{Back}_t^k \\ + s_{BackSuper}\text{Back Super}_t^k + Controls + \varpi_t^k\end{aligned} \tag{3}$$

where c_0 and d_0 are constants, and η_t^k and ϖ_t^k are the econometric errors.[19]

The key variable in equations (2) and (3) is IB_t^k, the installed base for console *k* at time *t*. In general, we expect the coefficients on this variable to be positive and significant. This captures the indirect network effect, as the installed base affects the supply of non-superstar and superstar games, which in turn affects hardware demand. We also interact installed base with console age to test our assertion that the effects of installed base on the supply of non-superstars and superstars varies as the platform ages. We account for backward compatibility in the software supply equations and include Back_t^k and Back Super_t^k as Kretschmer and Claussen (2016) show that the number of software units sold for the relevant previous generation console impacts the supply of software for backward compatible consoles. We expect Back_t^k and Back Super_t^k to negatively impact both non-superstar and superstar games supply.

Finally, we include the same control variables previously used in equation (1) in the software supply equations (equation (2) and equation (3)) to control for industry-wide and console-specific effects. We display variable definitions in Table 3.

-- Insert Table 3 about here --

[19] We use the levels of the number of available superstar and non-superstar games as the dependent variables in equations (2) and (3) instead of the natural log as suggested by Gretz (2010) to be consistent with how these variables enter the hardware demand specification in equation (1). Similarly, we use the level of installed base rather than the natural log. However, our results are qualitatively similar if we use a log-log or linear-log specification for software supply.

Identification and Endogeneity

There are three features of our modelling approach that allow us to identify the all of the parameters in equations (1), (2), and (3). First, and most importantly, we do not impose cross equation restrictions that require equations (2) and (3) to identify equation (1). Rather, we specify instruments for each endogenous variable in equation (1) to achieve identification. This approach to identification of hardware demand is consistent with previous literature that estimates both hardware demand and software supply (Clements and Ohashi 2005; Corts and Lederman 2009; Gretz 2010; Gretz and Basuroy 2013; Kim et al. 2014; Kretschmer and Claussen 2016; Nair et al. 2004). Second, the dependent variable in equation (1), $\ln\left(\text{Share}_t^k\right) - \ln\left(\text{Share}_t^0\right)$, does not enter as an independent variable in either equation (2) or equation (3) – rather the main independent variable is IB_t^k, the cumulative console sales prior to period *t*. Third, note that in equation (2) and equation (3) NonSuper_t^k is not a function of Super_t^k (i.e. Super_t^k is not an independent variable in equation (2)) and Super_t^k is not a function of NonSuper_t^k (i.e. NonSuper_t^k is not an independent variable in equation (3)), respectively. The second and third points together mean that there are no restrictions required in equations (1) or (3) to identify equation (2); similarly, there are no restrictions required in equations (1) or (2) to identify equation (3). This means the system is identified as long as each equation is identified individually (Wooldridge 2010). We use additional instruments to address endogeneity concerns and ensure identification in each equation.

We have several endogeneity concerns in the hardware demand equation and software supply equations. First, console prices are likely endogenous (equation (1)). The error term captures the effect of unobserved characteristics, which are likely considered when managers set console prices (Clements and Ohashi 2005; Zhu and Zhang 2010). Also, the number of available

superstar and non-superstar games may depend on the number of new consumers interested in the product. Clements and Ohashi (2005) show this to be likely if errors are autocorrelated.[20] Second, there is possible correlation between the installed base and the dependent variables in equation (2) and (3). This occurs as changes in the current period's installed base are likely related to software shocks in previous periods. We employ several popular cost-side instruments to address these concerns (see Dubé et al. (2010), Gretz & Basuroy (2013), and Kretschmer and Claussen (2016) for recent papers using similar instruments). We discuss the endogeneity issues and instruments in greater detail in Appendix A.

Results

Table 4 and Table 5 show descriptive statistics for hardware demand and software supply variables respectively and Table 6 shows correlations.

-- Insert Table 4 about here --

-- Insert Table 5 about here --

-- Insert Table 6 about here --

Note that the correlations of some of our key independent variables are high (e.g. the correlation between the number of superstars and non-superstar games is 0.899). We calculate variance inflation factors (VIFs), and find that in the hardware demand estimation and software supply estimations the mean VIFs are 4.51 and 2.69, respectively, and no individual VIF is above 10.[21] Given this, we conclude that multicollinearity is not a major concern.

Our estimation results are depicted in Table 7 through Table 9. In each table, we include a single equation Generalized Method of Moments (GMM) as well as joint (i.e. multi-equation) GMM estimation of equations (1) – (3). Our analysis relies on the joint GMM estimation shown

[20] We treat the number of backward compatible superstar and non-superstar games as exogenous, since they are more likely influenced by the market share of new buyers for the previous generation console they were originally designed for rather than the backward compatible, next-generation console.
[21] VIF tables are available from the authors upon request.

in columns (3) and (4) in Table 7, columns (7) and (8) in Table 8, and columns (11) and (12) in Table 9, because diagnostic tests show that joint GMM is superior to Ordinary Least Squares (OLS) and GMM on single equations. We provide more details of these tests in Appendix A. We estimate hardware demand and software supply equations together to obtain the joint GMM estimations.[22]

-- Insert Table 7 about here --

We first discuss hardware demand estimations in Table 7. Column (3) shows results without the interactions with console age while column (4) shows results with the interactions. Adding the interactions of non-superstar games and console age, as well as superstars and console age, significantly increases the explanatory power of the estimation.[23] In both the models, superstars have a positive and significant effect on market share (0.231, $p < 0.05$ and 0.685, $p < 0.05$ respectively). Also, the effect of superstars on hardware demand is larger than that of non-superstars.[24] The results in column (3) are in line with previous literature (Binken and Stremersch 2009; Corts and Lederman 2009; Kim et al. 2014) as we find that superstars matter while non-superstars do not. More importantly, as shown in column (4), both the hypothesized interaction terms are in the expected direction and statistically significant. In support of H1, we find that the interaction between superstars and console age is -.009 ($p < 0.05$) suggesting that as the console ages, the impact of superstars on hardware demand decreases. In support of H2, we find that the interaction between non-superstars and console age is .001 ($p < 0.01$) suggesting

[22] We cluster on console and use the optimal cluster robust weighing matrix to obtain the single equation GMM estimations. A similar cluster-robust weighing matrix is not feasible in the joint GMM estimation because there are more moments in the system than clusters (i.e. the cluster robust weighing matrix is not invertible). Instead we use a heteroskedastic and autocorrelation consistent weighing matrix with the lag order optimally selected using the Newey and West (1994) algorithm. Standard errors robust to clustering at the console level are presented in every estimation.

[23] Restricting the coefficients on the interaction terms in column (4) to zero yields $\chi^2(2) = 10.54$, $p < 0.01$ implying that the model with interactions has a higher explanatory power.

[24] Testing restrictions in columns (3) and (4) that the effect of superstars and non-superstar games are equal yields $\chi^2(1) = 4.06$, $p < 0.05$ for column (3) and $\chi^2(1) = 5.92$, $p < 0.05$ for column (4). Thus, in both cases we reject the null hypothesis that superstar and non-superstar games have equivalent effects on console market share.

that as the console ages, the impact of non-superstars on hardware demand increases. Thus, both H1 and H2 are supported. Further, the changing impact of superstars and non-superstar games as the platform ages differs.[25]

There are several other noteworthy findings in Table 7. In column (4), the main effects of non-superstar and superstar games are interpreted as the marginal impacts when console age is 0. Our estimates suggest the marginal effect of additional superstars becomes zero when the console is roughly 6.34 years old (.685/0.009 ≈ 76 months). On the other hand, the effect of non-superstar games is *negative* early on. Too many non-superstar games relative to superstars may signal poor quality. However, the impact of non-superstar games grows as consoles mature and becomes positive after the console is roughly 4 years old (0.048/0.001 = 48 months). The marginal impact of non-superstar and superstar games is equal at roughly 6.11 years[26], thereafter non-superstar games have a greater marginal impact on console adoption. Thus, while superstars are most important when a console is launched, non-superstar games are most important later, as they help extend a console's life. Figure 4 shows this pattern.

-- Insert Figure 4 about here --

Next, we discuss the results of superstars and non-superstar supply estimations from Table 8 and Table 9. We find that the impact of installed base on the provision of non-superstar and superstar games changes as the console ages. Including the interaction of installed base with console age significantly increases the explanatory power of the estimation.[27] Also, the marginal effect of installed base on the number of non-superstar games (63.270, $p < 0.01$) is greater than

[25] Restricting the coefficients on the interaction terms in estimation (4) to be the same yields $\chi^2(1) = 4.22$, $p < 0.05$ implying that the two interaction terms do not have the same magnitude.

[26] This is found by solving $0.685 - 0.009 \times \text{Console Age}_t^k = -0.048 + 0.001 \times \text{Console Age}_t^k$.

[27] A joint test of the hypothesis that the interaction terms are zero in column (8) in Table 8 and column (12) in Table 9 yields $\chi^2(2) = 184.13$, $p < 0.01$.

the impact on superstars (1.81, $p < 0.01$).[28] This is not surprising given that superstars are rare compared to the non-superstar games. The negative and significant interaction effects of installed base and console age on number of non-superstar games (-0.436, $p < 0.01$) and number of superstar games (-0.016, $p < 0.01$) indicate that the marginal impact of installed base on the provision of superstar and non-superstar games declines as the console gets older. The rates of decline for each type of game are significantly different ($\chi^2(1) = 169.64$, $p < 0.01$). However, it is important to note that these are absolute, not relative effects, and thus are not directly comparable.

-- Insert Table 8 about here --

-- Insert Table 9 about here --

To aid in interpretation we present the elasticity of non-superstar game supply to installed base and elasticity of superstars supply to installed base given by

$$\frac{\%\Delta \text{NonSuper}_t^k}{\%\Delta \text{IB}_t^k} = \frac{\partial \text{NonSuper}_t^k}{\partial \text{IB}_t^k} \times \frac{\text{IB}_t^k}{\text{NonSuper}_t^k} = \frac{\text{IB}_t^k}{\text{NonSuper}_t^k}\left(63.270 - 0.436 \times \text{Console Age}_t^k\right) \text{ and}$$

$$\frac{\%\Delta \text{Super}_t^k}{\%\Delta \text{IB}_t^k} = \frac{\partial \text{Super}_t^k}{\partial \text{IB}_t^k} \times \frac{\text{IB}_t^k}{\text{Super}_t^k} = \frac{\text{IB}_t^k}{\text{Super}_t^k}\left(1.810 - 0.016 \times \text{Console Age}_t^k\right) \text{ in Table 10.}$$[29] The formulas for the elasticities are derived using the coefficients on the levels and interaction terms in columns (8) and (12). We then calculate the elasticities at average variable values for the respective console age.[30]

There are several interesting findings in Table 10. First, the supply of non-superstar games is significantly more sensitive to installed base size than the supply of superstars. This

[28] A test of the hypothesis that the coefficients on IB_t^k are equal in column (8) in Table 8 and column (12) in Table 9 yields $\chi^2(1) = 156.53$, $p < 0.01$.

[29] These elasticities are calculated from the linear models of equations (2) and (3). We do not calculate elasticities for the non-linear model presented in equation (1) given the risk of misinterpretation (Ai and Norton 2003) and note this as a limitation of our current level of analysis.

[30] We note that these elasticities could be quite different if we use different variable values (Ai and Norton 2003).

holds throughout the hardware lifecycle.[31] Second, the elasticity of non-superstar games stays relatively constant at a little over 1 as hardware ages. This means that as the impact of non-superstar games on hardware demand becomes stronger, the innate attraction of hardware installed base to non-superstar games supply does not wane. Third, in contrast to non-superstar games, the elasticity of superstars to installed base has an inverted-U shape throughout the hardware lifecycle. The innate attraction of hardware installed base is weakest for superstars early in the life of hardware, when superstars have the greatest impact on hardware adoption, and as hardware is nearing the end of its life. The elasticity is largest, maxing out at a little over 0.6, when the impact of superstars on hardware demand is relatively low and approaching zero. As we discuss in the Managerial Insights section, these results have significant implications for hardware providers determining what type of software to invest in, and how much to invest, over the hardware lifecycle.

-- Insert Table 10 about here --

General Discussion

We identify hardware age as a hitherto unexplored moderator of the impact of the amount of available superstar and non-superstar software on hardware demand. Analysis of a large video game dataset using a joint GMM estimation shows that (1) the marginal impact of the amount of available non-superstars on hardware demand strengthens as hardware ages, (2) the marginal impact of the amount of available superstars weakens as hardware ages, and (3) eventually the amount of available non-superstars has a larger influence on hardware demand than the amount of available superstars. When we do not include the moderating effect of hardware age, we find that superstars impact hardware demand while non-superstars do not. These results are in line

[31] We also test to see if elasticities are the same over time within the same software type (separately for both superstars and non-superstar games). We reject the null hypothesis of similar coefficients in every case with $p < 0.01$. We suppress the test statistics given the number of individual tests conducted. The results are available from authors upon request.

with the previous research (Binken and Stremersch 2009; Corts and Lederman 2009; Kim et al. 2014). Including the moderating impact of hardware age obtains several important results regarding the impact of superstar and non-superstar software on hardware demand that did not have empirical support in the previous literature.

Additionally, by jointly modeling the software supply equations we observe that the impact of installed base on software supply is also moderated by hardware age. Specifically, we find that (4) the supply of non-superstar software is significantly more sensitive to installed base size than the supply of superstars, (5) the elasticity of non-superstar software to installed base stays relatively constant as hardware ages, and (6) the elasticity of superstars to installed base has an inverted-U shape throughout the hardware lifecycle.

In general, our findings speak to the strength of the indirect network effect, or the feedback loop whereby greater numbers of hardware adopters results in more software which attracts more hardware adopters, over the hardware lifecycle. Most of the empirical literature shown in Table 1 estimates the strength of indirect network effect, but none have considered how this strength is moderated by the hardware lifecycle. We contribute to the selected literature that explores absolute and relative differences in the strength of the indirect network effect. For example, Shankar and Bayus (2003) find differences in the strength of indirect network effects among firms within the same industry and Gretz and Basuroy (2013) show that the relative strengths of the indirect network effect compared to hardware quality effects change over the hardware generation lifecycle.

We find the indirect network is relatively weak early in the hardware lifecycle compared to later. This is because the impact of hardware installed base on superstar software supply is relatively weak when hardware is new – at precisely the time when superstar software has the largest impact on hardware demand. Conversely, the impact of hardware installed base on non-

superstar software supply remains relatively constant as hardware ages while the effect of non-superstars on hardware demand becomes stronger.

By identifying the age of hardware as a critical moderator of the effects of the amount of available superstar and non-superstar software on hardware demand, we highlight that in different phases of its lifecycle, hardware attracts different types of consumers. We argue that consumer heterogeneity, in turn, leads to different preferences for superstar and non-superstar software and related hardware demand. The failure to account for heterogeneity among the earlier and later hardware adopters will likely lead to over or underestimation of the impact of software types on hardware demand.

Changes in consumer preference for simplicity and price sensitivity over the product lifecycle are concepts that are applicable to other industries and other settings as well. Of course, the relative importance of superstars and non-superstars over the hardware lifecycle will depend on the preferences of earlier and later adopters, as well as the characteristics of superstars and non-superstars, in the particular industry. Our results suggest that it is important for theory to consider the preferences of these different adopter segments, as well as the key characteristics of superstars relative to non-superstars. While our focus is limited to the particular product strategy, our results highlight that it is important for theory to consider the preferences of adopter segments over the hardware lifecycle when examining other marketing strategies (e.g. price, advertising, and distribution) in platform and two-sided markets.

Managerial Insights

Our findings suggest strategies for hardware managers interacting with software firms (B2B) and customers (B2C). We present Figure 5 to summarize these strategies at the hardware-software and hardware-customer levels in the early and later phases of the hardware lifecycle. These strategies are driven by our findings concerning (1) the changing influence of the number

of available superstars and non-superstars on hardware demand over the hardware lifecycle, (2) the impact of installed base on the supply of superstars and non-superstars over the hardware lifecycle, and (3) our theorizing concerning customer heterogeneity and the preferences of early hardware adopters compared to later hardware adopters.

-- Insert Figure 5 about here --

During hardware introduction, relationships between hardware manufacturers and software developers are key to ensure superstar software availability. Our empirical analysis suggests that managers may *underinvest* in superstar provision when hardware is new if they do not account for the changing impact of superstars over the hardware's lifecycle. From Table 7, the coefficients on *Super* are 0.221 and 0.685 in column 3 and column 4 respectively. The predicted impact of superstar software on hardware demand at the time of hardware introduction is 3.10 (≈ 0.685/0.221) times greater when using the empirical model which accounts for the moderating effect of hardware age. This means when hardware is launched managers will grossly underestimate the impact of an additional superstar on hardware demand if they assume the marginal impact of superstar software on hardware adoption does not change over the hardware lifecycle. Importantly, hardware managers must recognize that the effect of installed base on superstar software supply is relatively weak. This finding, coupled with the impact of superstar software during hardware launch, should motivate hardware managers to form strong partnerships with software developers by offering creative arrangements including bundle offers, coupons, shared advertisement, pre-arranged orders, and other forms of cost subsidization.

However, the relatively weak impact of installed base on superstar software supply early in the hardware lifecycle is good news for later hardware entrants. Our findings suggest that a hardware firm that builds up an early installed base lead will be limited in its ability to leverage that position to attract relatively more superstars compared to an up-and-coming hardware

manufacturer who is new to the market. Both are in a similar relative bargaining position for superstar software since the influence of installed base on software supply is limited early in the lifecycle.

For hardware-customer relationships, our findings can help managers create appropriate advertising strategies highlighting the product's quality and innovativeness, as well as the technical capabilities that would ensure early adopters a full enjoyment of superstar software. At the retail level, product display and allocation should emphasize new hardware capabilities along with the catalog of superstar software. Any bundle of hardware and software should include superstar software. Additionally, managers should embrace both online and traditional distribution strategies. Finally, premium pricing can be utilized as early adopters are less price sensitive and have greater value for both hardware and associated superstars.

As hardware ages, managerial strategies should change accordingly. Hardware-software strategic relationships should adapt to ensure non-superstar software availability as this would help extend the life of an aging hardware, a prime concern for product and retail managers (Marchand 2016). While newer products can leverage the benefits of novelty, established products need to leverage different characteristics to make themselves desirable in the eyes of consumers. Common wisdom would encourage hardware managers to expand the portfolio of superstars to incentivize hardware adoption and prolong the life of hardware in the later stages of the lifecycle. Anecdotally, this is what appears to happen in the video game industry. Table 11 shows the average percentage of new games that are superstars on each console by the age of the console. Hardware managers always have an assortment of superstars and non-superstars on the console. However, our results suggest that hardware managers could increase adoption by increasing the rate of non-superstar introduction relative to superstar introduction as a console ages. Early on, managers follow this formula – the percentage of new games introduced that are

superstars initially falls after a console's introduction. However, hardware managers appear to increase the percentage of superstars introduced after a console's 5th year on the market. Our results speak against such a strategy. As later adopters prefer software that is simple to use, less engrossing, and inexpensive, we suggest managers should instead focus on increasing supply of non-superstar software as hardware ages. As an added bonus, this strategy seems relatively less costly for software developers as the development of superstar software is resource-intensive (Coughlan 2001).

-- Insert Table 11 about here --

In terms of cost subsidies, superstar supply becomes more elastic with respect to hardware installed base initially as hardware ages. This elasticity is greatest at around the same time that the marginal impact of superstars on hardware demand approaches zero. Superstar providers are most attracted to established hardware at the time when superstars have little influence on hardware demand. Therefore, hardware managers would be wise to limit financial and technical incentives for the provision of superstars and increase incentives for the provision of simpler and less expensive non-superstars as hardware enters the decline phase of its lifecycle. As a current example in the video gaming industry, software publishers often use Metacritic scores when determining bonuses of software developers. This practice gives explicit incentive to write superstar software rather than non-superstar software. To increase the supply of non-superstar games later in the hardware lifecycle, hardware managers can provide counter incentives that are independent of Metacritic scores and encourage creating simple and inexpensive games, rather than complex and expensive superstars.

Finally, hardware-customer managerial approaches should include advertising strategies that highlight an expansive non-superstar software library, product simplicity and lower prices. These tactics should mirror retail strategies that allocate and display products in a manner that

exemplifies the ample amount of non-superstar software available for the aging hardware. Given customer heterogeneities, we also suggest leveraging distribution strategies for hardware and software that are consistent with preferences for later adopters as the hardware enters the decline phase. For example, if later adopters prefer brick and mortar retailers to online retailers, traditional retailers could benefit by increasing the supply and selection of non-superstars in their stores.

Limitations and Directions for Future Research

In this section we discuss limitations of the current study and possibilities to further extend this research. First, although the theoretical arguments that we present may apply to multiple contexts, empirically we test the theory only using video games data. Future research can extend this to other two-sided markets (e.g. smartphones and apps; on-demand content providers (e.g. Netflix) and content; computer operating systems and software). Measuring the relative impact of price sensitivity, complexity, and risk aversion of adopters over the hardware lifecycle in different settings may also yield interesting findings. Second, incorporating consumer heterogeneity is much needed in the study of system products (R. S. Lee 2013; Rietveld and Eggers 2018; Steiner et al. 2016; Sun et al. 2016). We focus on heterogeneity with respect to price sensitivity and desire for software complexity, but other factors should be explored. This opens some exciting research opportunities. For example, individual and cultural differences regulate the amount of risk consumers are willing to take on (Avnet and Higgins 2006; Song et al. 2017). Managers and consumers often interpret the adoption of new technology as a risky endeavor. However, different strategies could minimize risk (e.g., peer reviews). This is yet to be tested empirically in the context of system markets.

Third, previous research shows that network externalities increase innovativeness (Stanko et al. 2007). While we do not model the effects of innovativeness in this study, its influence could be consistent with our findings. Particularly among early-adopters who value innovativeness (Chatterjee and Eliashberg 1990), their desire to maximize the utility derived from original software may also require a pairing with innovative hardware to fully exploit their mutual benefits. This, however, remains to be tested. We list more limitations in Appendix C.

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APPENDIX A – Endogeneity and Instruments

This study utilizes cost side instrumental variables to address endogeneity in all three of the estimation equations. Cost side instruments are appropriate to address endogeneity issues in a variety of demand estimations (see Nevo 2000) and similar instruments have been employed in several studies within our context (see Clements and Ohashi 2005; Dubé et al. 2010; Gretz and Basuroy 2013; Kretschmer and Claussen 2016). The key identification assumption is that cost shifters are likely taken into account by managers when choosing strategies such as price or number of available games but are likely not related to unobservable variables that impact consumer decisions. We discuss why we believe these assumptions hold for each of our instruments in each of our estimations below. We begin by describing the endogeneity issues and instruments used in the hardware demand equation then we discuss the software supply equations.

There are two main endogeneity issues in the hardware demand equation. First, according to Zhu and Zhang (2010), price is set by managers who likely use information not available to the researcher. Given this, price is likely correlated with the error term in the estimation. Second, new console adopters likely influence the number of superstar and non-superstar games available on a console. This results in a reverse causality issue – the availability of non-superstar games and superstars influences market share, but market share also influences the number of these two types of games available (Clements and Ohashi 2005).

To address these issues, we employ a series of cost side instruments used in other studies of the video game industry when dealing with similar endogeneity problems (Dubé et al. 2010; Gretz and Basuroy 2013). We use producer price indexes for Game Software Publishing (GSP), Computer Storage Device Manufacturing (CSDM), Electronic Computer Manufacturing (ECM), Magnetic and Optical Recording Media Manufacturing (MORMM), and Audio and Video

Equipment Manufacturing (AVEM) obtained from the U.S. Bureau of Labor Statistics (www.bls.gov) as cost shifters to identify console price as well as the number of superstar and non-superstar games. These instruments are relevant (i.e. likely correlated with the endogenous regressors) because (1) GSP is related to the costs of game publishing while (2) CSDM, ECM, MORMM, and AVEM reflect the average cost of components used in console manufacturing and game manufacturing. We believe these are valid instruments (i.e. unrelated to the error term in the estimation) because these industry wide measures of costs are likely calculated not taking into account unobservable variables related to hardware demand. As noted by Dubé et al. (2010, pg. 226), "PPI's provide sources of exogenous variation in prices that are excluded from demand and that are unlikely to be correlated with consumer tastes for video game consoles".

We use both the level and natural log of GSP to allow for possible non-linear effects of game publication cost shocks on console prices and game provision. The use of GSP to identify console price exploits the two-sided nature of the video game market. As Rysman (2009, pages 129 – 130) notes "prices on both sides of the market depend on the joint set of demand elasticities and marginal costs on each side...The low price on one side not only attracts elastic consumers on that side but also, as a result, leads to higher prices or more participation on the other side". Given this we expect managers to raise (reduce) console price in response to exogenous cost shocks that decrease (increase) game prices.

The impact of GSP on the number of superstar or non-superstar games is more complex. First, we expect a positive shock to the cost of publishing games to result in fewer superstar and non-superstar games. However, because of the two-sided nature of the market, an increase in software prices will likely be compensated by a reduction in console price, resulting in a larger number of adopters and a larger market for games (Rysman 2009). This second effect would offset to some extent the impact of the first effect. Further, we expect the second effect is

augmented if a positive change in GSP is reflective of an increase in cost due to higher average software quality. Since first-stage estimations are reduced form, the relationship of GSP to both type of game's provision may be either negative (if the first effect dominates) or positive (if the second effect dominates).

The other instruments (CSDM, ECM, MORMM, and AVEM) are correlated with console and game manufacturing costs. In general, we expect an increase (decrease) in console manufacturing costs to prompt managers to increase (decrease) price to cover costs; we expect an increase (decrease) in game manufacturing costs to decrease (increase) the number of games (both superstars and non-superstars) managers provide to the market. However, as we highlight above, the complex relationship between costs and prices on both sides of a two-sided market (Rysman 2009) may moderate our predicted effects.

In addition to the level and log of GSP, we use the change in CSDM and ECM from month to month to capture the impact of changes in manufacturing costs for both consoles and games. These changes are likely positively correlated with console price and negatively correlated with the number of superstar and non-superstar games.

Alone, GSP, CSDM, and ECM capture industry wide cost shocks. However, Rossi (2014) highlights that PPIs alone may not be sufficient instruments because they do not vary by cross-sectional unit, they only vary over time. We create console specific instruments by interacting console age with the other PPIs, MORMM and AVEM. Using different PPIs to create the interaction terms adds additional variation to the first-stage estimation to help identify the endogenous regressors and reduces the concern of collinearity of the interaction terms with the PPIs used to capture industry wide cost shocks. Interacting MORMM and AVEM with console age is appropriate because manufacturing processes and costs change throughout the life of a console (Coughlan 2001). We expect cost shocks to have a smaller effect on prices over time –

cost shocks are more easily absorbed later on when managers have adopted more efficient production processes and can leverage their greater experience. This should manifest itself in negative coefficients on the interaction terms in the first-stage estimation for console price. In contrast we expect positive coefficients on these terms in the first-stage estimations for superstar and non-superstar games (and these types of games interacted with console age). Recall greater manufacturing costs of games likely leads to fewer games (a negative coefficient for the non-interacted PPI variable) – a positive coefficient on the interaction term of the PPI with console age means this initial negative impact is diminished as the console matures.[32] We also include the interaction of console age with CSDM as an additional IV to help separately identify the number of available non-superstar games, the number of available superstars, and their interaction terms with console age in estimations (2) and (4).

As a final set of instruments to identify price and the number of superstar and non-superstar games, we leverage average game age which is the average age of software titles available on the console in the period. Software title age is measured in months since the game was initially introduced on the respective console. It is likely that consoles with older games on average command a relatively lower price; it is likely that consoles with older games have to increase the number of new non-superstars and superstars available to fill missing supply (Kretschmer and Claussen 2016). However, while other studies of the video game industry have used average age of software titles on the console as an instrument to identify price and the number of available games (Clements and Ohashi 2005; Gretz 2010; Kretschmer and Claussen 2016), we are concerned that average game age may not satisfy the exclusion restrictions if software developers release games taking into account forecasts of unobservable variables that

[32] We note that our results do not change qualitatively if we change which PPIs we interact with console age and which PPIs we first difference to capture changes in manufacturing costs. However, the first-stage F-stats of other specifications are not as large as those presented here.

impact hardware demand.[33] To overcome this concern, we instead use the mean of average game age from competing consoles in the same console generation. For example, as an instrument for console *k* in period *t*, we use the mean of the average game age for all other consoles in the same console generation as console *k* in period *t*. This is similar to using average of competitor characteristics as exogenous instruments as suggested by Nevo (2000). The key identifying assumption is that developers planning their release dates for their games on other consoles (not console *k*) do not systematic take into account future unobservables that impact hardware demand for the focal console (console *k*). Germann, Ebbes, & Grewal (2015) employ this logic as well when obtaining instruments derived from competitor characteristics for their study on the impact of CMOs; more recently, Malshe, Colicev, & Mittal (2017) derive instruments for customer satisfaction of a focal firm using characteristics of competing firms. In order for the instrument to be invalid (i.e. correlated with the unobservable), all competitors would have to simultaneously take coordinated action against the focal firm based on unobservable variables of the focal firm. We believe this is highly improbably in our setting, especially given the long lead time, often several years (Coughlan 2001), necessary to develop games. Both other console makers and game developers for other consoles would have to coordinate, taking strategic action – timing their release of all games on their consoles – based on an accurate forecast (given the long lead time necessary to develop games) of a future unobservable of a competing console in order for the orthogonality condition of this instrument to be violated. While this is highly unlikely, as an additional safeguard we use the change in mean of average software age on competing consoles from month to month as the instrument rather than the level – violation of the exclusion restriction here requires the added complexity of other consoles and the game

[33] We thank an anonymous reviewer for pointing this out.

developers for other consoles coordinating their release times based on accurate forecasts of two consecutive periods of unobservables that impact hardware demand for the focal console.

However, we expect mean of average game age on competing consoles to impact price and the number of available non-superstars and superstars on the focal console. We expect a positive relationship between the change in mean of average game age on competing consoles and console price as consoles makers may have to lower their price to compete with consoles that have newer games on average. Similarly, we expect a negative relationship between the change in mean of average game age on competing consoles and the supply of non-superstars and superstars as console makers will face pressure to increase game supply to better compete with competitors that have a newer catalog of games available.

Also, we interact mean of average game age on competing consoles with GSP as we expect higher software publishing costs will moderate the main effect on console price and the number of available non-superstars and superstars. We use the change in this variable from month to month as the instrument to be consistent with our transformation of the mean of average game age on competing consoles. First, higher game prices (reflected by higher GSP) reduce the amount that console makers will have to lower console price to compete with consoles with a newer game catalog. As such, we expect the interaction between of GSP and the mean of average game age on competing consoles to have a negative impact on price – decreasing the positive impact of the main effect on price. Second, we expect higher software publishing costs to moderate the rate at which game developers can introduce new software to replace old supply. Given this, we expect a positive coefficient on the interaction of GSP and the mean of average game age on competing consoles in the availability of non-superstar and superstar games' estimations – moderating the negative impact of the main effect on the supply of these two types of games.

We further test the validity of change in the mean of average game age on competing consoles and its interaction with GSP by computing the GMM distance statistic which compares the difference in the Hansen's J of the model with all instruments to the Hansen's J of the model without the instruments being tested (Hayashi 2000). Under the null hypothesis, all instruments are exogenous and the test statistic is distributed as a chi-square with degrees of freedom equal to the number of instruments being tested; under the alternative hypothesis, the suspect instruments are correlated with the econometric error term in the second stage estimation and thus not valid. The key assumption of this test statistic is that all other instruments besides those being tested are exogenous. We believe this is highly likely since all other instruments are created from exogenous cost shifters and interactions with console age. The GMM distance statistic is $\chi^2(2) = 0.064$ with a p-value = 0.969 – we fail to reject the null which gives us additional confidence that the instruments created from the mean of average game age of competing consoles are exogenous.

In the software supply equations, we are concerned that shocks in software supply during previous periods may influence the installed base of the current period (Clements and Ohashi 2005). We leverage the two-sided nature of the market to obtain appropriate instruments. As with the hardware demand equation, we note that as games become more expensive managers may have an incentive to lower console price in order to encourage adoption (Rysman 2009). Again, we use exogenous costs shifter likely to be reflected in software price – AVEM and GSP. We use the change is GSP from month to month as an instrument and expect it to be positively related to installed base as managers have an incentive to lower console price (to encourage more adoption) as games become more expensive when they are costlier to produce. Second, we use the natural log of AVEM to capture the impact of industry wide exogenous changes in prices on adoption – we expect a negative impact as higher prices likely result in a smaller installed base.

Third, we create a console specific instrument by interacting the natural log of AVEM with console age.[34] We expect a positive coefficient on the interaction term since cost shocks are likely more easily absorbed by older consoles as the production process tends to be more efficient due to greater experience. For our fourth instrument, we leverage competition on the console side of the market. With more competition, an individual console is likely to have fewer adopters since consumers have more choices which may result in a decrease in the installed base. We use the number of other consoles in the same console generation as the focal console as an instrument to capture competitive intensity. Lastly, as in the hardware demand equation, we use the change in the mean of average game age on competing consoles. We expect that console installed base will be larger if competitors have older games on average. Again, we test the validity of a change in the mean of average game age on competing consoles using the GMM distance statistic (Hayashi 2000) in both the superstar and non-superstar games supply equations. The resulting test statistics are $\chi^2(1) = 1.844$ and $\chi^2(1) = 0.800$, respectively, with p-values = 0.175 and 0.371 – we fail to reject the hypothesis that the instrument is orthogonal in both estimations.

In general, we expect the predicted effects of the instruments on installed base to have the opposite effect on installed base interacted with console age – the impact of lower console prices on adoption is likely to be more important early on rather than later, when the console is established. However, we expect a negative impact of the number of competitors – consoles become less attractive as they age, as a result competition is likely to have a larger negative effect on the adoption rate of older consoles.

[34] We note that we obtain similar results if we use the level of AVEM (along with the level of AVEM interacted with console age) rather than the natural log of AVEM, however the first-stage F-stats are higher using natural log of AVEM.

We show the results of our first-stage estimations in Table A.1 and Table A.2. All first-stage F-statistics are well over 10 indicating the instruments are strong (Wooldridge 2010). Also, the signs of the instruments when they are significant are consistent with our expectations outlined above; by and large the signs of the instruments when they are not significant are also consistent with our expectations. Additionally, each instrument is significant in at least one first-stage estimation. Finally, the overidentification restrictions are satisfied in every estimation suggesting the instruments are orthogonal to the error term.

-- Insert Table A.1 about here --

-- Insert Table A.2 about here --

Diagnostic tests for GMM Superiority

First, the Durbin-Wu-Hausman test (Wooldridge 2010) is statistically significant when comparing OLS and single equation GMM estimations for hardware demand and software supply equations, suggesting endogeneity is present.[35] Therefore, OLS estimates are likely biased. Second, GMM estimates are preferred if the instruments are exogenous and correlated with the endogenous variables. The non-significant Hansen J-statistics in all the GMM estimations supports the former condition, while the large (over 10) and significant first-stage F-stats support the latter (Rossi 2014; Stock and Watson 2007; Stock and Yogo 2002; Wooldridge 2010). Third, GMM estimation is more efficient than traditional Two-Stage Least Squares estimation (or Three-Stage Least Squares in the case of joint estimation) in the presence of arbitrary heteroskedasticity (Stock et al. 2002). Lastly, joint estimation of hardware demand and

[35] For the hardware demand estimations, the Durbin-Wu-Hausman test comparing OLS and column (1) in Table 7 yields $\chi^2(3) = 14.15$, $p < 0.01$ while comparing OLS and column (2) yields $\chi^2(5) = 15.65$, $p < 0.01$. Similarly, for the non-superstar supply estimations, the Durbin-Wu-Hausman test comparing OLS and column (5) results in $\chi^2(1) = 147.38$, $p < 0.01$ while comparing OLS and (6) yields $\chi^2(2) = 86.62$, $p < 0.01$. For the superstar supply estimations, the Durbin-Wu-Hausman test comparing OLS and column (9) yields $\chi^2(1) = 138.29$, $p < 0.01$ while comparing OLS and column (10) yields $\chi^2(2) = 119.33$, $p < 0.01$.

superstar/non-superstar software supply improves efficiency if errors are correlated across equations.

APPENDIX B – Analysis with Exclusive Games/Non-Exclusive Games

Previous research in the video game industry has shown that exclusive games and non-exclusive games impact hardware demand differently (Kim et al. 2014; R. S. Lee 2013). Similarly, the impact of installed base on the supply of exclusive games and the supply of non-exclusive games is shown to be significantly different (Corts and Lederman 2009). In this appendix we report the analysis using only exclusive games and using only non-exclusive games in order to see if (1) the main results in the paper hold for these different game types and (2) to see if there are significant differences in our estimations based on game type.

Our main analysis for both exclusive and non-exclusive games are in Tables B.2 – B.4 where results for Hardware Demand, Number of Non-Superstar Games, and Number of Superstar Games are displayed, respectively. We report estimations using only exclusive games for each console in the column titled "Using only Exclusive Games." Here, we redefine NonSuper_t^k, Super_t^k, $\text{NonSuper}_t^k \times \text{Console Age}_t^k$, $\text{Super}_t^k \times \text{Console Age}_t^k$, Back_t^k, and Back Super_t^k using only games that are exclusive to the respective console. Similarly, we redefine the variables using only games that are non-exclusive to the respective console and report estimations in the column titled "Using only Non-Exclusive Games." Both estimations are joint GMM estimations; the same column title in each table indicates the results are from the same joint GMM estimation including the 3 equations. Descriptive statistics for the redefined variables are displayed in Table B.1 under their respective estimation title.

-- Insert Table B1 about here --

-- Insert Table B2 about here --

-- Insert Table B3 about here --

-- Insert Table B4 about here --

Importantly, the signs and significance of the main independent variables in both estimations are consistent with the main findings in the paper – our main results in the paper hold if we only consider exclusive games or non-exclusive games instead of all games in the estimation equations. However, our analysis suggests that the effects of exclusive and non-exclusive games are significantly different – we provide a variety of coefficient tests comparing both estimations in Table B.5.

-- Insert Table B5 about here --

We performed 6 tests to see if the coefficient estimates on the main independent variables in the "Using only Exclusive Games" and "Using only Non-Exclusive Games" models are significantly different. Each column in Table B.5 represents a Wald test where the coefficients on the variables indicated by 'X' are constrained to be equal in both the "Using only Exclusive Games" and the "Using only Non-Exclusive Games" estimation. For example, the column titled "Test 2: Main Independent Variables in the Hardware Demand equation" is a joint test of significant differences between

(1) the coefficient on NonSuper_t^k in the Hardware Demand equation in the "Using only Exclusive Games" estimation and the coefficient on NonSuper_t^k in the Hardware Demand equation in the "Using only Non-Exclusive Games" estimation;

(2) the coefficient on $\text{NonSuper}_t^k \times \text{Console Age}_t^k$ in the Hardware Demand equation in the "Using only Exclusive Games" and the coefficient on $\text{NonSuper}_t^k \times \text{Console Age}_t^k$ in the Hardware Demand equation in the "Using only Non-Exclusive Games" estimation;

(3) the coefficient on Super_t^k in the Hardware Demand equation in the "Using only Exclusive Games" estimation and coefficient on Super_t^k in the Hardware Demand equation in the "Using only Non-Exclusive Games" estimation;

(4) the coefficient on $\text{Super}_t^k \times \text{Console Age}_t^k$ in the Hardware Demand equation in the "Using only Exclusive Games" and the coefficient on $\text{Super}_t^k \times \text{Console Age}_t^k$ in the Hardware Demand equation in the "Using only Non-Exclusive Games" estimation.

We note that we reject the null hypothesis of statistically similar coefficients with at least 90% confidence for each test.

Test 1 is a general test of similar coefficients of the main independent variables in the Hardware Demand, Number of Non-Superstar Games, and Number of Superstar Games equations. Results suggest that the impacts of the variables are significantly different between models. Test 2 specifically considers the Hardware Demand equation. Results suggest that the impact of exclusive and non-exclusive games on hardware demand are significantly different. Test 3 considers only non-superstars. We find that the impact of exclusive non-superstars on hardware demand is significantly different than the impact of non-exclusive non-superstars. Comparing coefficients, it appears that the impact of exclusive games is larger (in absolute value) than non-exclusive games. Similarly, Test 4 considers only superstars; results suggest that exclusive superstars have a significantly larger impact on hardware demand than exclusive non-superstars. Test 5 and Test 6 focus on the software supply equations. Test 5 shows that console installed base has a significantly larger impact on the supply of exclusive non-superstar games compared to non-exclusive non-superstar games. Similarly, Test 6 shows that the same is true for superstars – console installed base has a significantly larger impact on the supply of exclusive superstars compared to non-exclusive superstars.

In sum, our findings are consistent with previous literature that has considered exclusive and non-exclusive games separately (Corts and Lederman 2009; Kim et al. 2014; R. S. Lee 2013). In general, exclusive games have a larger (in absolute terms) impact on hardware demand than non-exclusive games (consistent with Kim et al. (2014) and Lee (2013)). This is true for both superstars and non-superstars. Also, exclusive software supply is more sensitive to installed base than non-exclusive software supply (consistent with Corts and Lederman (2009)) – again, this finding is consistent for both superstar and non-superstar supply.

APPENDIX C – FURTHER LIMITATIONS

Future research should also consider differences in the utility derived from hedonic (e.g., video games) vs utilitarian (e.g., computers) products. We argue that the phase of the hardware lifecycle will moderate the impact of different software types on hardware sales in either setting. However, this moderation depends on the characteristics of superstars and non-superstars in relation to preferences of earlier hardware adopters compared to later adopters within the industry setting.[36]

We note that we do consider the complexities associated with relating hardware adoption to expectations of future software (superstar or non-superstar) availability. Expectations of the number of future software available and hardware success may have an influence on current hardware adoption (Steiner et al. 2016). This is especially important when considering the role of superstars in launching new hardware to compete against an established incumbent. Research has shown that anticipated regret may influence willingness to purchase in a variety of settings (Zeelenberg 1999) especially when technology is fast evolving (Shih and Schau 2011). We expect this to have a limited impact in our setting where new generations of video game consoles are introduced every 5 to 7 years, but may be more relevant in industries like smartphones which usually have yearly introductions of new hardware. However, understanding how quality signals of superstars may influence anticipated regret or heuristics early adopters may employ when making decisions early in the hardware lifecycle is a useful area of future research in platform and two-sided markets.[37]

Also, we do not consider possible inefficiencies caused by the compensation strategies we recommend to encourage superstar and non-superstar development over the hardware

[36] We thank an anonymous reviewer for this suggestion.
[37] We thank an anonymous reviewer for this suggestion.

lifecycle. Software developers and hardware managers often have exclusive deals in the video game industry (e.g. 70.7% of games are released exclusively to a single console in our sample) which means our strategy suggestions can be implemented by hardware managers in negotiation with exclusive software providers. However, there may be inefficiencies or inequities that could occur for software developers that serve multiple hardware firms with the same game – especially when each hardware firm would prefer a different level of game quality[38]. Speaking to these possible inefficiencies, or potential remedies, is beyond the scope of the current paper and we leave it to future research.

Despite these limitations, our findings make significant strides by establishing boundary conditions on previous findings in the literature – both superstars and non-superstars matter, but at different point in the hardware lifecycle. For product managers and retailers, these insights offer a compelling way to evaluate product introduction and allocation strategies, as our results show that a positive financial impact could be derived from a more insightful evaluation of product characteristics in two-sided markets.

[38] We thank an anonymous reviewer for pointing out this issue.

Figure 1
Hardware Lifecycle as a Moderator of Indirect Network Effects of Superstar and Non-Superstar Software Supply

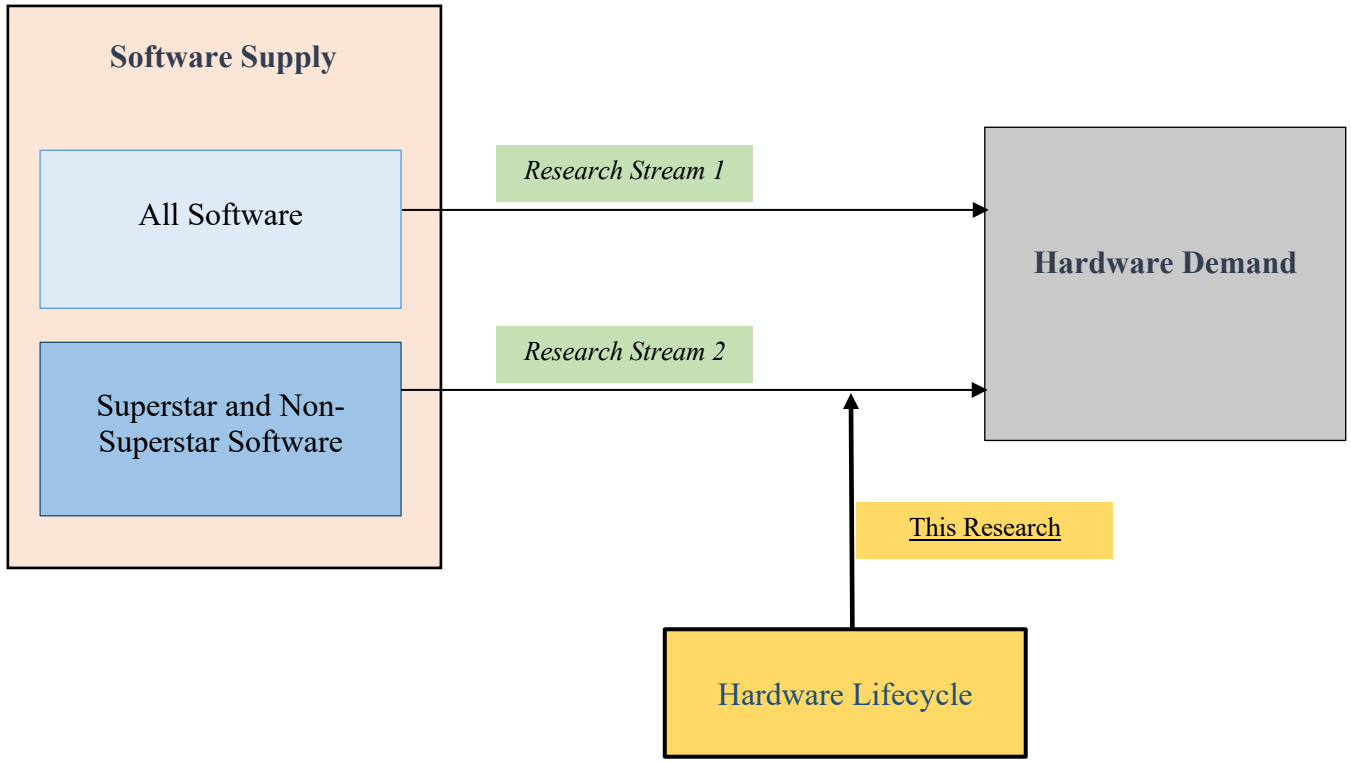


Figure 2
Price Differences between Superstar and Non-Superstar Games

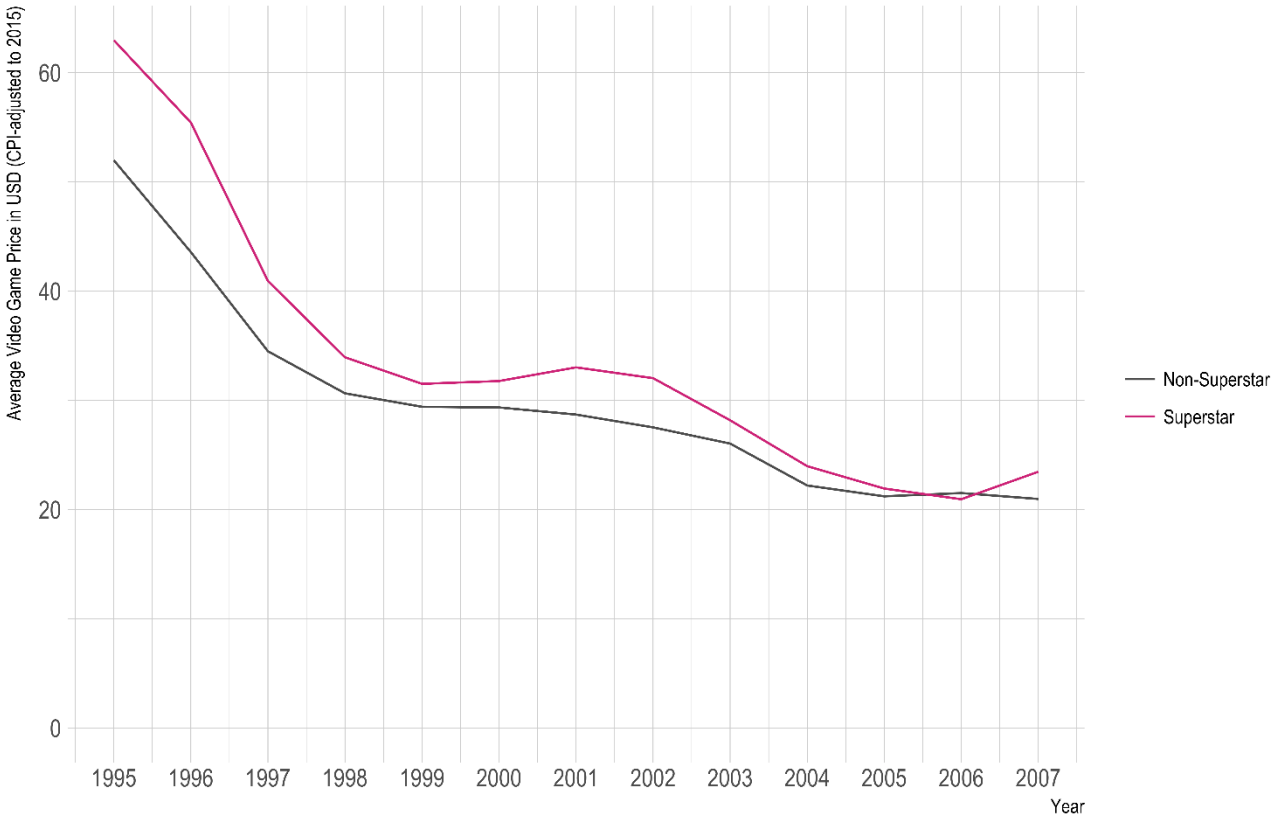


Figure 3
Disproportionately Large Sales Revenues of Superstar Video Games

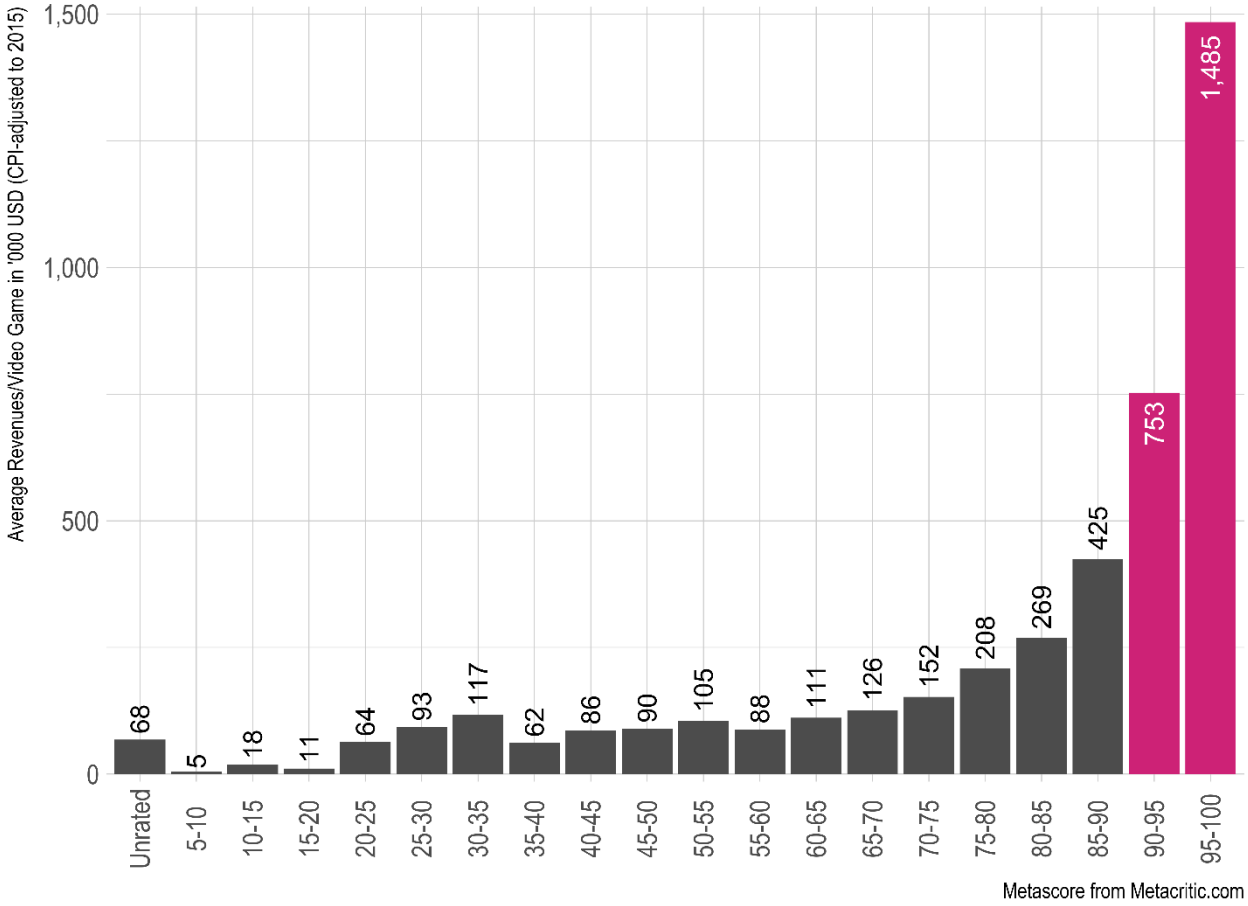


Figure 4
Different Impacts of Superstars and Non-Superstars on Hardware Demand

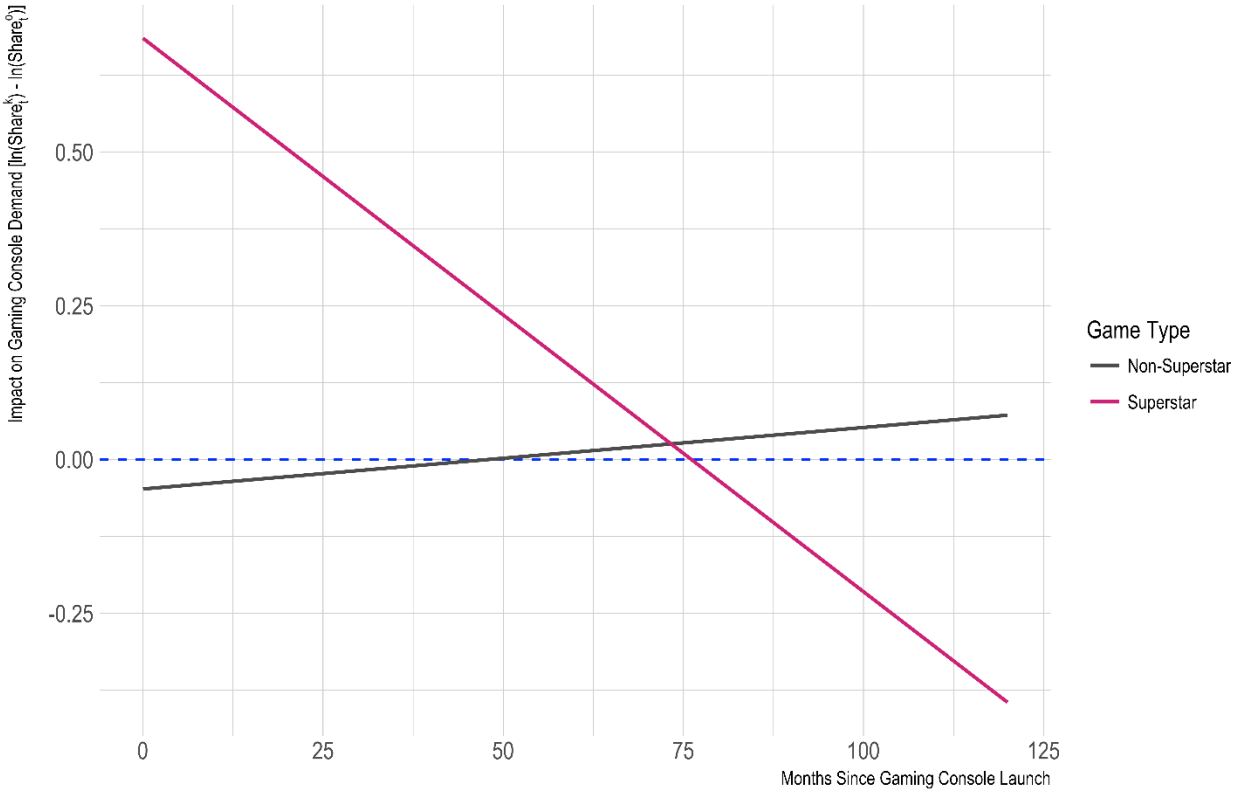


Figure 5
Managerial Strategies Related to the Impact of Hardware Lifecycles on Hardware Demand and Software Supply

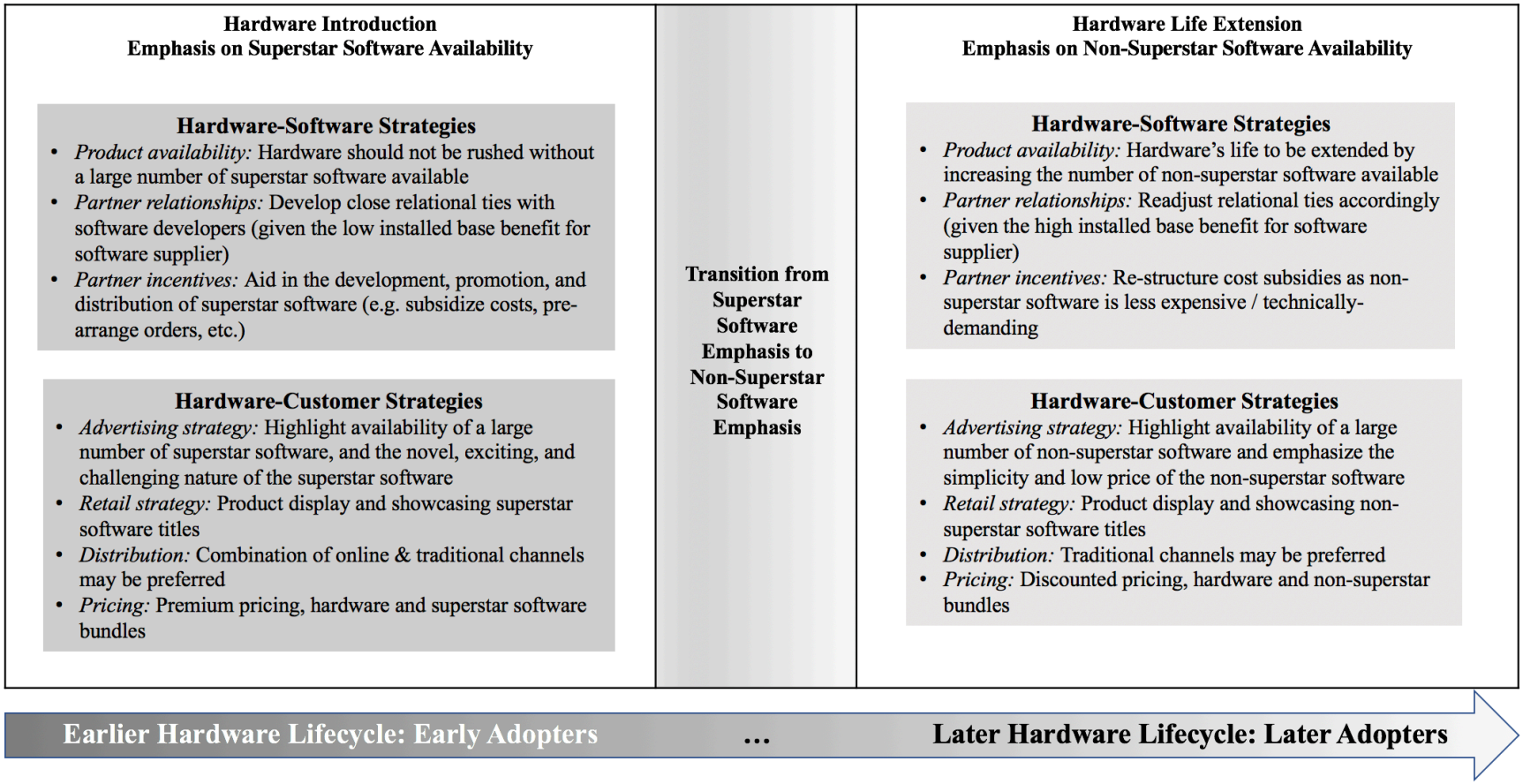


Table 1. Literature review.
Summary of illustrative research based on software quality versus availability and hardware lifecycle.

Reference	Context	Feedback Loop Analysis	Consideration of Hardware Lifecycle	Research Focus: (software quality vs availability)	Key Results
Stremersch et al. (2007)	Various industries (e.g., CD, DVD, video games, laser discs)	Model hardware demand and software supply	Focus on product introduction Includes time trend as a control variable	Diminishing effects of indirect network effects	Product availability and indirect network effects produce weaker-than-expected results.
Binken and Stremersch (2009)	Video games	Model hardware demand	Includes product age as a control variable	Strengthening effects of software quality	Superstar software exerts a disproportionally large effect on hardware unit sales.
Corts and Lederman (2009)	Video games	Model hardware demand and software supply	Includes product age as a control variable	Strengthening effects of software quality	Software 'hits' (e.g., high quality software) has a positive and significant effect on hardware demand and perceived utility.
Kim et al. (2014)	Video games	Model hardware demand and software supply	Includes product age as a control variable	Strengthening effects of software quality	"Only high-quality software offerings increase hardware sales" (p. 100)
Kretschmer and Claussen (2016)	Video games	Model hardware demand and software supply	Includes product age as a moderator for the impact of backward compatible software	Strengthening effects of software availability -- specifically backward compatible software	Access to backward compatible software increases hardware adoption. This effect diminishes with console age.
Nair et al. (2004)	Personal digital assistant	Model hardware demand and software supply	No hardware lifecycle effects considered	Strengthening effects of software availability	Indirect network effects explain about 22% of log-odds ratio of the sales of different PDA's
Clements and Ohashi (2005)	Video games	Model hardware demand and software supply	Includes product age as control variable and instrument	Strengthening effects of software availability	Penetration pricing is important early in the life of the console, whereas installed base is important later on.

Dubé et al. (2010)	Video games	Model hardware demand and software supply	Includes product age as a control variable	Strengthening effects of software availability	Indirect network effects can lead to a strong and economically significant increase in market share dominance.
Gretz and Basuroy (2013)	Video games	Model hardware demand and software supply	Hardware lifecycle effects considered	Strengthening effects of software availability	Indirect network effects and hardware quality play an important role on product adoption.
Gandal et al. (2000)	Compact disks	Model hardware demand and software supply	No hardware lifecycle effects considered	Strengthening effects of software availability	"A 10% increase in CD titles would have as large an effect as a 5% [hardware] price cut" (p. 57)
Dranove and Gandal (2003)	Digital video disc	Model hardware demand	No hardware lifecycle effects considered	Strengthening effects of software availability	Indirect network effects play a positive and significant effect in the DVD market and serves as an entry barrier for alternative products.
This project	**Video games**	**Model hardware demand and software supply**	**Hardware lifecycle effects considered**	**Strengthening effects of both, software quality and availability**	*** The importance of superstar and non-superstar software is moderated by the hardware lifecycle.** *** Non-superstar software is important in the later stages of the hardware's life.** *** Superstar software is important in the earlier stages of the hardware's life.**

Table 2. Console generations and console characteristics.

Console Gen.	Consoles	Release Date	Processing Unit Size	Average RAM*	Game Media	*N* in the dataset	Number of Games Released for the Console		
							All Games	Exclusive Games	% Exclusive Games
1	Sega Genesis	Aug 1989	16-Bit	0.322	Cartridge and Disk	81	683	385	56.37%
	Super Nintendo	Aug 1991				75	694	401	57.78%
	Sega CDX	Apr 1994				51	148	85	57.43%
2	3DO	Oct 1993	32-Bit and 64-Bit	3.108	Cartridge and Disk	43	158	97	61.39%
	Atari Jaguar	Nov 1993				39	45	25	55.56%
	Sega Saturn	May 1995				62	251	101	40.24%
	Sony PlayStation	Sep 1995				146	1353	969	71.62%
	Nintendo 64	Sep 1996				103	290	170	58.62%
3	Sega Dreamcast	Sep 1999	128-Bit	42.250	Disk and DVD	46	250	140	56.00%
	Sony PlayStation 2	Oct 2000				85	1639	698	42.59%
	Microsoft Xbox	Nov 2001				72	911	243	26.67%
	Nintendo GameCube	Nov 2001				72	574	134	23.34%
4	Microsoft Xbox 360	Nov 2005	Multiple Processors	371.667	Disk, DVD, HD DVD, and Blu-ray	24	213	54	25.35%
	Nintendo Wii	Nov 2006				12	132	49	37.12%
	Sony PlayStation 3	Nov 2006				12	83	15	18.07%
					Total:	**923**	**7424**	**3566**	
					Avg. % Exclusive Games on Each Console				**45.88%**

*Average Random Access Memory (RAM) Contained in a Console in the Product Generation (in Megabytes)

Table 3. Variable definitions.

Variable	Definition
Share_t^k	Market share of console k in period t.
Share_t^0	Market share of the outside option in period t.
NonSuper_t^k	The number of non-superstar games available for console k in period t.
Super_t^k	The number of superstar games available for console k in period t.
Back_t^k	The cumulative number of games released for previous generation console compatible with console k by period t.
Back Super_t^k	The share of superstar games out of the cumulative number of games released for previous generation console compatible with console k by period t.
IB_t^k	Cumulative monthly console sales prior to period t.
Price_t^k	Average console price in period t. Prices are adjusted for inflation using the Consumer Price Index for Urban Consumers (1980 – 1982 = 100)
Console Age_t^k	The number of months at time t since console k was introduced to the market.

Table 4. Descriptive statistics for hardware demand estimations.

	Mean	Std. Dev.	Min.	Max.
$\ln\left(\text{Share}_t^k\right) - \ln\left(\text{Share}_t^0\right)$	-7.477	3.250	-16.875	-2.145
Price_t^k (in \$100s)	0.778	0.567	0.021	2.955
NonSuper_t^k	310.550	293.797	4.000	1364.000
Super_t^k	13.644	10.958	0.000	51.000
Back_t^k	221.210	509.233	0.000	2992.000
Back Super_t^k	0.007	0.015	0.000	0.051
Console Age_t^k (CA)	51.727	35.795	1.000	146.000
°GSP	44.451	33.857	0.000	103.300
°* Δ Computer Storage Device Manufacturing PPI_t (CSDM)	-1.885	3.085	-19.600	7.200
°* Δ Electronic Computer Manufacturing PPI_t (ECM)	-3.376	3.454	-23.700	0.100
°* Δ Mean of Average Game Age on Competing Consoles $_t^k$ (GA)	0.263	3.184	-71.201	4.764

$N = 923$.
* $N = 908$.
° Instruments used in estimation
Δ ≡ change in variable from period $t - 1$ to period t.

Table 5. Descriptive statistics for number of non-superstars and number of superstars estimations.

	Mean	Std. Dev.	Min.	Max.
NonSuper_t^k	310.550	293.797	4.000	1364.000
Super_t^k	13.644	10.958	0.000	51.000
IB_t^k (1,000,000s)	11.707	10.316	0.000	38.800
Back_t^k	221.210	509.233	0.000	2992.000
Back Super_t^k	0.007	0.015	0.000	0.051
Console Age_t^k (CA)	51.727	35.795	1.000	146.000
°No. of Competing Consoles in the Same Generation as Console k_t^k (Comps)	2.015	1.025	0.000	4.000
°* Δ Mean of Average Game Age on Competing Consoles $_t^k$ (GA)	0.263	3.184	-71.201	4.764
°* Δ Game Software Publishing PPI_t (GSP)	-0.404	2.340	-18.400	6.400
°ln(Audio and Video Equipment Manufacturing PPI_t (AVEM))	4.322	0.072	4.160	4.432

$N = 923$.
* $N = 908$.
° Instruments used in estimation
Δ ≡ change in variable from period $t - 1$ to period t.

Table 6. Correlations.

	$Share_t^k$	$Price_t^k$	$NonSuper_t^k$	$Super_t^k$	$Back_t^k$	$Back\ Super_t^k$	$Console\ Age_t^k$
$Share_t^k$	1.000						
$Price_t^k$	0.295	1.000					
$NonSuper_t^k$	0.324	-0.215	1.000				
$Super_t^k$	0.243	-0.296	0.899	1.000			
$Back_t^k$	0.323	0.456	0.183	0.134	1.000		
$Back\ Super_t^k$	0.169	0.371	-0.031	-0.082	0.733	1.000	
$Console\ Age_t^k$	-0.366	-0.696	0.300	0.361	-0.237	-0.285	1.000
IB_t^k	0.090	-0.475	0.698	0.663	0.029	-0.173	0.793

$N = 923$

Table 7. Hardware demand estimations.

	GMM Estimations		Joint GMM Estimations	
	(1)	(2)	(3)[a]	(4)[b]
$Price_t^k$	-1.281***	-4.046***	-1.873*	-3.112**
	(0.382)	(1.264)	(1.045)	(1.325)
$NonSuper_t^k$	-0.008*	-0.054**	-0.006	-0.048***
	(0.004)	(0.021)	(0.005)	(0.015)
$NonSuper_t^k \times Console\ Age_t^k$		0.001***		0.001***
		(0.000)		(0.000)
$Super_t^k$	0.277***	0.821***	0.221**	0.685**
	(0.102)	(0.280)	(0.109)	(0.289)
$Super_t^k \times Console\ Age_t^k$		-0.011**		-0.009**
		(0.004)		(0.005)
$Back_t^k$	0.008	0.007	0.004	0.011
	(0.005)	(0.011)	(0.007)	(0.008)
$Back\ Super_t^k$	-270.682***	-751.763***	-332.927***	-687.364**
	(61.991)	(255.258)	(115.781)	(295.287)
[c] Controls	✓	✓	✓	✓
N	908	908	908	908
J Statistic	2.765	2.890	18.119	6.749
1-stg F-Stats:				
$Price_t^k$	26.78***	18.39***		
$NonSuper_t^k$	51.42***	33.62***		
$NonSuper_t^k \times Console\ Age_t^k$		22.31***		
$Super_t^k$	27.93***	203.09***		
$Super_t^k \times Console\ Age_t^k$		42.10***		

Dependent variable: The natural log of console share subtracted from the natural log of outside option share.
*, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.
(Standard errors robust to console level clustering presented in parenthesis.)
[a] Estimated jointly with estimation (7) in Table 8 and estimation (11) in Table 9.
[b] Estimated jointly with estimation (8) in Table 8 and estimation (12) in Table 9.
[c] Controls include month dummies, year dummies, hardware dummies, and console age. Coefficients on control variables are not displayed for brevity.
Instruments for GMM and joint GMM estimations are listed in Table 4.

Table 8. Number of non-superstar games estimations.

	GMM Estimations		Joint GMM Estimations	
	(5)	(6)	(7)[a]	(8)[b]
IB_t^k	69.321***	66.318***	67.244***	63.270***
	(10.565)	(5.675)	(4.063)	(4.969)
$IB_t^k \times \text{Console Age}_t^k$		-0.461***		-0.436***
		(0.043)		(0.034)
Back_t^k	-0.883	-0.522*	-0.999***	-0.302
	(0.736)	(0.281)	(0.262)	(0.261)
Back Super_t^k	26982.040***	2394.705	25797.150***	362.431
	(7702.253)	(5345.036)	(4954.061)	(4833.188)
[c] Controls	✓	✓	✓	✓
N	908	908	908	908
J Statistic	6.374	2.147	18.119	6.749
1-stg F-Stats:				
IB_t^k	24.59***	24.59***		
$IB_t^k \times \text{Console Age}_t^k$		28.72***		

Dependent variable: number of regular (non-superstar) games.
*, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.
(Standard errors robust to console level clustering presented in parenthesis.)
[a] Estimated jointly with estimation (3) in Table 7 and estimation (11) in Table 9.
[b] Estimated jointly with estimation (4) in Table 7 and estimation (12) in Table 9.
[c] Controls include month dummies, year dummies, hardware dummies, and console age. Coefficients on control variables are not displayed for brevity.
Instruments for GMM and joint GMM estimations are listed in Table 5.

Table 9. Number of superstar games estimations.

	GMM Estimations		Joint GMM Estimations	
	(9)	(10)	(11)[a]	(12)[b]
IB_t^k	2.298***	1.945***	2.158***	1.810***
	(0.224)	(0.218)	(0.154)	(0.174)
$IB_t^k \times$ Console Age_t^k		-0.016***		-0.016***
		(0.002)		(0.002)
$Back_t^k$	-0.034*	-0.008	-0.033**	-0.000
	(0.020)	(0.014)	(0.013)	(0.012)
Back $Super_t^k$	521.080**	-346.729	470.618**	-535.544**
	(241.727)	(274.180)	(229.087)	(243.552)
[c] Controls	✓	✓	✓	✓
N	908	908	908	908
Adj. R^2				
J Statistic	5.505	2.871	18.119	6.749
1-stg F-Stats:				
IB_t^k	24.59***	24.59***		
$IB_t^k \times$ Console Age_t^k		28.72***		

Dependent variable: number of superstar games.
*, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.
(Standard errors robust to console level clustering presented in parenthesis.)
[a] Estimated jointly with estimation (3) in Table 7 and estimation (7) in Table 8.
[b] Estimated jointly with estimation (4) in Table 7 and estimation (8) in Table 8.
[c] Controls include month dummies, year dummies, hardware dummies, and console age. Coefficients on control variables are not displayed for brevity.
Instruments for GMM and joint GMM estimations are listed in Table 5.

Table 10. Elasticity of non-superstar and superstar supply to installed base.

	1 Month After Hardware Introduction	1 Year After Hardware Introduction	3 Years After Hardware Introduction	5 Years After Hardware Introduction	7 Years After Hardware Introduction	9 Years After Hardware Introduction
*Elasticity of $NonSuper_t^k$ with respect to IB_t^k	1.196	1.149	1.061	1.064	1.096	1.063
*Elasticity of $Super_t^k$ with respect to IB_t^k	0.397	0.520	0.580	0.634	0.494	0.101
χ^2 (p-value) for hypothesis test that Elasticity of $NonSuper_t^k$ with respect to IB_t^k = Elasticity of $Super_t^k$ with respect to IB_t^k	148.31 (< 0.01)	56.74 (< 0.01)	32.99 (< 0.01)	10.32 (< 0.01)	10.13 (< 0.01)	31.50 (< 0.01)

* Elasticities calculated using average variable values for $NonSuper_t^k$, $Super_t^k$, and IB_t^k at the console age indicated in the column.

Hypotheses tests that Elasticities are equal to each other within software type (e.g. Elasticity of $NonSuper_t^k$ with respect to IB_t^k at 1 Month After Hardware Introduction = Elasticity of $NonSuper_t^k$ with respect to IB_t^k at 1 Year After Hardware Introduction) are all rejected with p-values < 0.000. Results of individual χ^2 tests are available from authors upon request.

Table 11. Average superstar introduction by console age

	Console less than a year old	Console between 1 and 2 years old	Console between 2 and 3 years old	Console between 3 and 4 years old	Console between 4 and 5 years old	Console between 5 and 6 years old	Console between 6 and 7 years old
Average % Games Introduced that are Superstars by Console	5.30%	4.40%	3.16%	2.93%	1.93%	2.44%	3.33%

Table A.1 First-stage regressions for hardware demand.

The dependent variable in each estimation is indicated across the first row					
	$Price_t^k$	$NonSuper_t^k$	$NonSuper_t^k \times$ Console Age_t^k	$Super_t^k$	$Super_t^k \times$ Console Age_t^k
Excluded Instruments					
ln(GSP)	0.249	345.957**	29759.170**	0.746	844.956
	(0.206)	(174.535)	(14305.730)	(7.631)	(533.054)
GSP	-0.006*	-3.583	-351.597	0.072	-7.561
	(0.003)	(3.754)	(282.026)	(0.152)	(10.187)
Δ CSDM	-0.002	-1.886**	-94.973	-0.086***	-3.688
	(0.002)	(0.837)	(68.095)	(0.030)	(2.250)
Δ ECM	-0.001	-2.399***	-176.229**	-0.068***	-4.639***
	(0.002)	(0.929)	(78.049)	(0.025)	(1.714)
Δ GA	0.00004**	-0.054**	-1.610	-0.0004	0.084
	(0.00002)	(0.027)	(1.968)	(0.0008)	(0.069)
Δ (GSP × GA)	-0.001	3.807**	192.311	0.046	-1.834
	(0.001)	(1.886)	(145.190)	(0.059)	(5.660)
MORMM × CA	-0.0003***	0.151	10.491	0.007**	0.542***
	(0.0001)	(0.128)	(11.305)	(0.004)	(0.208)
AVEM × CA	0.0001	0.800***	65.068**	0.026***	1.940**
	(0.0002)	(0.207)	(27.071)	(0.006)	(0.767)
Control Variables					
CSDM × CA	-0.0001***	0.046***	1.340	0.001***	0.005
	(0.0000)	(0.017)	(0.872)	(0.000)	(0.023)
$Back_t^k$	-0.002***	1.462***	79.008***	0.042***	2.097**
	(0.001)	(0.319)	(27.785)	(0.013)	(1.059)
$Back\ Super_t^k$	-39.754**	-43607.780***	-2156951.0***	-1740.590***	-81408.510***
	(18.778)	(6422.109)	(408215.500)	(236.155)	(17343.840)
[a] Controls	✓	✓	✓	✓	✓
R^2	0.857	0.699	0.594	0.764	0.739
Partial R^2 of Excl. Instrum.	0.337	0.530	0.309	0.564	0.298
1st Stage F-Statistics on IVs	18.39***	33.62***	22.31***	203.09***	42.10***

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.
$N = 908$.
(Standard errors robust to console level clustering presented in parenthesis.)
[a] Controls include month dummies, year dummies, hardware dummies, and console age.

Table A.2 First-stage regressions for software supply estimations.

The dependent variable in each estimation is indicated across the first row

	IB_t^k	$IB_t^k \times \text{Console Age}_t^k$
Excluded Instruments		
Comps	-1.819***	-154.496***
	(0.343)	(28.447)
Δ GA	0.041***	2.683
	(0.011)	(1.820)
Δ GSP	-0.025	-6.078*
	(0.028)	(3.169)
ln(AVEM)	-64.070*	7883.295***
	(37.294)	(2096.266)
ln(AVEM) × CA	1.011***	-113.149***
	(0.352)	(35.447)
Control Variables		
Back_t^k	0.043***	1.649**
	(0.007)	(0.773)
Back Super_t^k	-1079.053***	-64600.320***
	(89.631)	(9124.252)
[a] Controls	✓	✓
R^2	0.871	0.923
Partial R^2 of Excl. Instrum.	0.284	0.349
1st Stage F-Statistics on IVs	24.59***	28.72***

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.
$N = 908$.
(Standard errors robust to console level clustering presented in parenthesis.)
[a] Controls include month dummies, year dummies, hardware dummies, and console age.

Table B.1 Descriptive Statistics for Additional Estimations with only Exclusive Games and only Non-Exclusive Games

Descriptive Statistics for Each Estimation Presented Under the Estimation Title.

	Mean	Std. Dev.	Min.	Max.
Using only Exclusive Games				
$NonSuper_t^k$	155.724	163.902	3.000	711.000
$NonSuper_t^k \times Console\ Age_t^k$	10059.452	13837.347	6.000	63600.000
$Super_t^k$	7.163	5.489	0.000	21.000
$Super_t^k \times Console\ Age_t^k$	465.933	507.754	0.000	2108.000
$Back_t^k$	120.481	279.267	0.000	969.000
$Back\ Super_t^k$	0.007	0.015	0.000	0.060
Using only Non-Exclusive Games				
$NonSuper_t^k$	158.513	161.619	1.000	804.000
$NonSuper_t^k \times Console\ Age_t^k$	9415.009	11333.322	2.000	68340.000
$Super_t^k$	6.660	6.945	0.000	32.000
$Super_t^k \times Console\ Age_t^k$	393.608	496.194	0.000	2604.000
$Back_t^k$	98.185	283.994	0.000	2294.000
$Back\ Super_t^k$	0.007	0.015	0.000	0.053

$N = 908$.

Descriptives displayed only for variables that changed from the models presented in Tables 7 – 9.

Table B.2 Hardware Demand Estimations with only Exclusive Games and only Non-Exclusive Games

	Using only Exclusive Games	Using only Non-Exclusive Games
$Price_t^k$	-6.946***	-3.554***
	(1.878)	(1.303)
$NonSuper_t^k$	-0.151***	-0.119***
	(0.056)	(0.037)
$NonSuper_t^k \times Console\ Age_t^k$	0.002***	0.002***
	(0.001)	(0.000)
$Super_t^k$	2.236**	1.085***
	(1.022)	(0.417)
$Super_t^k \times Console\ Age_t^k$	-0.025**	-0.022***
	(0.012)	(0.008)
$Back_t^k$	-0.004	0.095*
	(0.022)	(0.048)
$Back\ Super_t^k$	-544.081	-502.072**
	(650.539)	(205.506)
[a] Controls	✓	✓
N	908	908
J Statistic	6.396	10.257
1-stg F-Stats:		
$Price_t^k$	19.590***	46.752***
$NonSuper_t^k$	51.141***	79.366***
$NonSuper_t^k \times Console\ Age_t^k$	18.408***	216.020***
$Super_t^k$	23.346***	22.239***
$Super_t^k \times Console\ Age_t^k$	19.723***	14.253***

Dependent variable: The natural log of console share subtracted from the natural log of outside option share.
*, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.
(Standard errors robust to console level clustering presented in parenthesis.)
[a] Controls include month dummies, year dummies, hardware dummies, and console age. Coefficients on control variables are not displayed for brevity.
All results are joint GMM. Results in each column are from joint estimations with their counterparts shown in Tables B.3 and B.4.

Table B.3 Number of Non-Superstar Games Estimations with only Exclusive Games and only Non-Exclusive Games

	Using only Exclusive Games	Using only Non-Exclusive Games
IB_t^k	34.006***	25.956***
	(5.648)	(1.913)
$IB_t^k \times$ Console Age_t^k	-0.199***	-0.247***
	(0.015)	(0.032)
$Back_t^k$	-0.132	-0.041
	(0.461)	(0.587)
Back $Super_t^k$	9961.397	-2903.752
	(10653.572)	(1931.656)
[a] Controls	✓	✓
N	908	908
J Statistic	6.396	10.257
1-stg F-Stats:		
IB_t^k	26.195***	23.901***
$IB_t^k \times$ Console Age_t^k	15.058***	19.089***

Dependent variable: number of non-superstar games.
*, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.
(Standard errors robust to console level clustering presented in parenthesis.)
[a] Controls include month dummies, year dummies, hardware dummies, and console age. Coefficients on control variables are not displayed for brevity.
All results are joint GMM. Results in each column are from joint estimations with their counterparts shown in Tables B.2 and B.4.

Table B.4 Number of Superstar Games Estimations with only Exclusive Games and only Non-Exclusive Games

	Using only Exclusive Games	Using only Non-Exclusive Games
IB_t^k	1.013***	0.723***
	(0.149)	(0.100)
$IB_t^k \times$ Console Age_t^k	-0.007***	-0.009***
	(0.001)	(0.002)
$Back_t^k$	0.001	0.036
	(0.017)	(0.031)
Back $Super_t^k$	368.101	-282.294***
	(355.068)	(100.949)
[a] Controls	✓	✓
N	908	908
J Statistic	6.396	10.257
1-stg F-Stats:		
IB_t^k	26.195***	23.901***
$IB_t^k \times$ Console Age_t^k	15.058***	19.089***

Dependent variable: number of superstar games.
*, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.
(Standard errors robust to console level clustering presented in parenthesis.)
[a] Controls include month dummies, year dummies, hardware dummies, and console age. Coefficients on control variables are not displayed for brevity.
All results are joint GMM. Results in each column are from joint estimations with their counterparts shown in Tables B.2 and B.3.

Table B.5 Tests of Significantly Different Coefficients Between "Using only Exclusive Games" and "Using only Non-Exclusive Games" Estimations in Tables B.2 – B.4.

'X' Indicates which Variables we Test to Have Statistically Similar Effects in Both Estimations.

Equation	Independent Variables	Test 1: Main Independent Variables in Hardware Demand, Number of Non-Superstar Games, and Number of Superstar Games equations	Test 2: Main Independent Variables in the Hardware Demand equation	Test 3: The Impact of non-superstars in the Hardware Demand equation	Test 4: The Impact of superstars in the Hardware Demand equation	Test 5: The impact of installed base on non-superstar supply in the Number of Non-Superstar Games equation	Test 6: The impact of installed base on superstar supply in the Number of Superstar Games equation
Hardware Demand	$NonSuper_t^k$	X	X	X			
	$NonSuper_t^k \times Console\ Age_t^k$	X	X	X			
	$Super_t^k$	X	X		X		
	$Super_t^k \times Console\ Age_t^k$	X	X		X		
Number of Non-Superstar Games	IB_t^k	X				X	
	$IB_t^k \times Console\ Age_t^k$	X				X	
Number of Superstar Games	IB_t^k	X					X
	$IB_t^k \times Console\ Age_t^k$	X					X
	χ^2 statistic	19.52**	11.79**	5.42*	8.76**	4.94*	8.62**
	Degrees of freedom	8	4	2	2	2	2
	p-value	0.012	0.019	0.067	0.013	0.085	0.014

*, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.